

UNION CHRISTIAN COLLEGE ALUVA

[Re-accredited with A++ grade (Vth cycle) by National Assessment and Accreditation council (NAAC)]



STUDENT PORTAL

REPORT OF MINIPROJECT

Submitted in partial fulfilment of the requirement for the award of the degree of

B.Sc. COMPUTER SCIENCE

By Mahatma Gandhi University, Kottayam during 2025 - 2026

Submitted by

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Under the guidance of ***Mrs. Elizabeth Thomas***

UNION CHRISTIAN COLLEGE ALUVA

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Department of Computer Science

CERTIFICATE

This is to certify that the Mini Project report entitled “***STUDENT PORTAL***” submitted by ***Dhony Thomas (Reg No: 2300210024477), Ebin Antu (Reg No: 2300210024478) and Ashlin V (Reg No: 210021027719)*** in partial fulfilment of the requirements for the award of BSc in Computer Science during the academic year 2025 - 2026.

The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Computer Science. The work is carried out as an in-house project by the student.

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Submitted for the university examination held on:

Internal Examiner

External Examiner

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ABSTRACT

Computers today have become indispensable tools. They are fast changing the way in which work is done by their speed, accuracy and diligence. Almost all good businesses practices have embraced computers for increasing the productivity and efficiency, which ultimately results in profits.

Information technology has created an interactive and expensive quality to information not experienced previously thus contributing to information not experienced previously thus contributing to information becoming a primary commodity. Information is no longer static. Instead, it can be continually added to growing in value the original data, message or idea is augmented. The impact of information technology is communication process and connections challenge basic assumptions about organization structure, working relationship and the nature and quality of service.

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1. INTRODUCTION

1.1 OVERVIEW OF THE SYSTEM

The *Student Portal* is a web-based application designed to simplify and centralize important academic information for students. In today's digital age, managing student records, fee details, and exam schedules manually can be time-consuming and prone to errors. The Student Portal aims to provide an easy-to-use platform where students can conveniently access all their academic details in one place. It serves as a bridge between students and the institution, promoting transparency and efficiency in academic management. The system is designed in such a way that no formal programming knowledge is expected from the user.

Accurate headings and guidelines are provided wherever necessary. The system is found to be effective and proves to be versatile as it's a very flexible with user-friendly screens enabling the user to use it without any inconvenience. This provides a better system, and it overcomes many problems that exist in the current system. This project focuses on providing key features such as viewing personal and academic details, checking fee structures and payment status, and accessing exam-related information like timetables and results notifications. By integrating these functionalities, the portal reduces administrative workload and ensures that students are always updated with accurate and real-time information. The design of the portal emphasizes user-friendliness and security to ensure that only authorized users can access sensitive data.

The project highlights the importance of digital solutions in modern education and provides a foundation for future enhancements such as online fee payment, academic notifications, and digital document management. It not only saves time and effort but also contributes to a more organized and transparent academic environment.

1.2 PROBLEM DEFINITION AND OBJECTIVE

1. Problem Definition: In many educational institutions, students often face challenges in accessing essential academic information such as their personal details, fee structure, and examination schedules. This information is usually maintained manually or across different systems, which leads to duplication of work, delays in updates, and communication gaps between students and administration. The absence of a centralized and digital platform makes it difficult for students to retrieve the latest information efficiently. As a result, manual data management not only consumes time and effort but also increases the chances of errors and data inconsistency.

2. Objectives: The main objective of the *Student Portal* system is to provide a centralized digital platform that allows students to conveniently access their academic information in a secure and efficient manner. The system is designed to automate various student-related processes, reduce administrative workload, and ensure that students receive accurate and up-to-date information. The portal aims to improve communication between students and the institution while maintaining transparency and reliability in data management. The platform aims to address the following key issues:

1. Information Gathering - The system collects and organizes essential student information such as personal details, fee structures, and examination data in a structured database. This ensures that all relevant information is easily accessible and can be updated by authorized users whenever necessary.

2. Efficient Coordination - The portal enables smooth coordination between students and the administration by serving as a common communication platform. It helps reduce manual interaction, minimizes paperwork, and ensures that both students and staff can perform their tasks efficiently.

3. Data Security and Privacy - The system incorporates authentication and access control mechanisms to protect sensitive student data from unauthorized access. Privacy and data integrity are maintained through secure login credentials and proper database management techniques.

4. Scalability - The *Student Portal* is designed to be scalable, allowing future enhancements such as online fee payment, attendance tracking, and performance monitoring. This ensures that the system can adapt to the growing needs of educational institutions and remain useful in the long term.

2. SYSTEM ANALYSIS

2.1 INTRODUCTION

System analysis is a detailed study of various operations performed by the system and their relationships within and outside the system. It is a process of gathering and interpreting facts, diagnosing problems and using this information to recommend improvements to the system. System analysis is an important activity that takes place when building new information systems or changing the existing ones. The systems are often made up of many interrelated tasks. Changes to many of these tasks or addition of new tasks can affect existing ones. The investigation into the system operation and possible changes to the system is called analysis. System analysis is the way of studying a system with an eye on solving its problem using computers. It is the most essential part of the project development. During analysis, data are collected on available files, decision points, and transactions handled the present system. Training experience and common sense are required for the collection of information needed to do the analysis. Once the analysis is completed, analysts have firm understanding of what is to be done. To analyze a system, one must study the system work in detail before designing the appropriate computer-based system that will meet all requirements of the system.

System analysis includes the following steps:

- Requirement analysis
- Feasibility study
- Analysis
- Design

Implementation

- Post implementation and maintenance

The above steps constitute a logical framework for system analysis.

2.2 EXISTING SYSTEM

The system consists of database that must be maintained with various information. The report generation and printing work are difficult. In the existing system every time a reference should be made. There are no high possibilities to commit errors and mistakes, which leads to produce the wrong statements to the management. Report keeping is also not an easy work.

Once the details are fed into the system there is no need for various persons to deal with separate section. Only one person is enough to maintain all the reports.

Another important drawback of existing system is time factor. It will not help the management to solve the problem in time. The management could not act on any issue; purchase of production unless getting the reports in time controls cannot be used.

Demerits of Existing System

- * Physical volume of the data is very large.
- * The delay in information search and retrieval.
- * Problems in updating and backup.
- * Damage of papers containing the information.
- * Less Efficiency and accuracy due to lot of manual entries
- * More Time Delay for payments.
- * Increased expenditure for storage

This is evident in the difficulty of tracing misplaced documents Cannot be able to prepare the invoice quickly.

2.3 PROPOSED SYSTEM

All the drawbacks involved in the manual system can easily be solved by the introduction of the computerized system. The Student Portal is

a web-based system designed to provide students with quick and easy access to all their academic and personal information. The proposed system overcomes the limitations of traditional, manual data management methods by offering a centralized platform where students can log in and view their fee structure, personal details, and examination information. It aims to replace paper-based processes with a digital system that ensures accuracy, efficiency, and transparency in managing student data.

The portal allows administrators to update or modify student details, manage fee records, and upload exam timetables or results in real-time. Students can then access this information securely from anywhere using their credentials. The system also ensures data consistency by maintaining all records in a structured database. Additionally, the design focuses on user-friendliness, allowing even non-technical users to navigate the system easily.

Through the proposed Student Portal, institutions can improve communication with students, minimize manual errors, and reduce the administrative burden. The system acts as a one-stop solution for student-related services and can be further extended with additional modules like online payments, attendance tracking, or notifications to enhance overall functionality.

Advantages of proposed system

The proposed Student Portal offers several advantages over the existing manual or semi-digital systems. Firstly, it saves time for both students and administrators by automating repetitive tasks and allowing quick retrieval of information. Students no longer need to visit different departments for fee, exam details or signature of HOD for the confirmation of students whether the student is their own departments and apply for their course certificates, as all information is available on a single online platform.

Secondly, the system improves accuracy and reduces the chances of human error. Since all information is stored in a centralized database, updates made by the administration are reflected instantly across the system. The portal also promotes transparency by ensuring that students can always access the most recent and correct data.

Another major advantage is the enhanced security of data. The system uses secure login and authentication mechanisms to ensure that only authorized users can access or modify information. It also minimizes paperwork and storage costs, supporting a more eco-friendly and efficient administrative process. The server will collect the data and store them only a database.

Scope of the proposed system

The Student Portal has a broad scope in the educational sector, as it can be implemented in schools, colleges, and universities to simplify student information management. Initially, the system focuses on providing features such as student profile management, fee structure details, and exam information. However, it has been designed with scalability in mind, allowing easy integration of new modules in the future

Benefits of proposed system

- Students can easily access their fee details, exam schedules, and personal information without visiting administrative offices. It reduces manual work and saves time for both students and staff.
- All student data is stored in a centralized database, minimizing errors and inconsistencies.
- The system is designed with simplicity in mind so that users can easily navigate and access the required information. Even non-technical users can operate the system efficiently.

- Secure login credentials and access control ensure that only authorized users can view or modify information.
- Students can view all relevant details themselves, reducing dependency on administrative staff.
- Reduces the use of paper forms and physical records, contributing to an eco-friendly environment.

2.3.1 MODULE DESCRIPTION

The project "Student Portal" is developed under the Windows 8 platform along with HTML and CSS as front end and PHP for server-side scripting and MySQL Server as database. This software includes mainly of three modules. They are as follows.

1. Admin module
2. Office module
3. Student module

The **Admin Module** is the core part of the system that allows the administrator to manage and control all other modules. The admin has the highest level of authority and is responsible for maintaining the overall functionality of the portal. This includes managing user accounts, assigning roles, and ensuring that data remains accurate and secure. The admin can add, update, or delete student and office records, manage database backups, and monitor system performance. Additionally, the admin has access to reports and statistics that provide insights into various academic and administrative activities.

The **Office Module** is designed for use by administrative or office staff who handle the day-to-day operations related to students. This module enables the office personnel to manage student information, fee details, and exam schedules efficiently. They can update student records, enter new data such as payment details or exam timetables, and generate reports when needed. The office module serves as the bridge between the admin and the students, ensuring that all updates made by the staff are reflected accurately in the student portal. It helps streamline communication and reduces manual paperwork.

The **Student Module** is the front-facing part of the system that allows students to log in securely and access their academic and personal information. Through this module, students can view their personal details, fee structure, payment status, examination schedules and apply for their course certificates. They can also check notifications, results, and other updates posted by the office or admin. This module is designed with a user-friendly interface so that students can easily navigate and retrieve the information they need without assistance. It improves accessibility, transparency, and communication between students and the institution.

2.4 FEASIBILITY STUDY

Before developing the *Student Portal* system, a feasibility study was conducted to analyse whether the project is practical and achievable with the available resources. The study helps determine if the system can be developed effectively and implemented successfully within the institution. The main types of feasibility considered for this project include **technical feasibility**, **economic feasibility**, and **operational feasibility**.

2.4.1 TECHNICAL FEASIBILITY:

- The system is developed on the Windows 8 platform using HTML, CSS, PHP, and MySQL — all open-source and reliable technologies.
- These technologies are easy to integrate, cost-free, and widely supported.
- The required hardware and software resources are easily available.
- The system is scalable and can be upgraded with new features in the future.

2.4.2 ECONOMIC FEASIBILITY:

- The project is cost-effective since it uses open-source tools that require no licensing fees.
- It reduces paper usage and administrative costs by automating manual processes.

- Maintenance and operational costs are minimal once the system is deployed.
- Provides long-term financial benefits by improving efficiency and saving time.

2.4.3 OPERATIONAL FEASIBILITY:

- The system is user-friendly, requiring minimal technical knowledge to operate.
- Students, office staff, and administrators can easily use their respective modules.
- Enhances coordination, accuracy, and communication within the institution.
- It can be smoothly integrated into the institution's existing workflow.

Conclusion:

The Student Portal provides students with easy access to academic information such as personal details, fee structure, exam schedules, online payment, and notifications. It improves administrative efficiency, ensures data accuracy, and enhances communication between students and the institution.

Recommendations:

- Implement the system across educational institutions for better student services.
- Add future enhancements such as attendance tracking and performance analytics.
- Expand notification features for alerts regarding exams, fees, and other important updates.

By conducting a comprehensive feasibility study, the Student Portal project can ensure informed decision-making, leading to successful implementation and enhanced management of student data and services.

2.5 SYSTEM SPECIFICATION

2.5.1 TOOLS/PLATFORM

FRONT END

1. HTML

Hypertext Markup Language (HTML) is the standard markup language for creating web pages and web applications. With Cascading Style Sheets (CSS) and JavaScript, it forms a triad of cornerstone technologies for the World Wide Web.

Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document.

HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as `<img>` and `<input>` directly introduce content into the page. Other tags such as `<p>` surround and provide information about document text and may include other tags as sub-elements. Browsers do not display HTML tags but use them to interpret the content of the page.

HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages. Inclusion of CSS defines the look and layout of content. The World Wide Web Consortium (W3C), maintainer of both the HTML and the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.

2. CSS

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language like HTML. CSS is a

cornerstone technology of the World Wide Web, alongside HTML and JavaScript.

CSS is designed to enable the separation of presentation and content, including layout, colors, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, and reduce complexity and repetition in the structural content.

Separation of formatting and content also makes it feasible to present the same markup page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille-based tactile devices. CSS also has rules for alternate formatting if the content is accessed on a mobile device.

The name cascading comes from the specified priority scheme to determine which style rule applies if more than one rule matches a particular element. This cascading priority scheme is predictable.

BACK END

1.PHP

PHP started out as a small open-source project that evolved as more and lots of people found out how useful it was. Rasmus Lerdorf unleashed the first version of PHP way back in 1994.

- PHP is a recursive acronym for 'PHP: Hypertext Pre Processor.
- PHP is a server-side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, session tracking, even build entire e-commerce sites.
- It is integrated with several popular databases, including MySQL, PostgreSQL, Oracle, Sybase, Informix, and Microsoft SQL Server. PHP is forgiving PHP language tries to be as forgiving as possible. PHP Syntax is C-Like. Common uses of PHP,

- PHP performs system functions, i.e. from files on a system it can create, open, read, write, and close them. PHP can handle forms, i.e. gather data from files, save data to a file, through email you can send data and return data to the user.
- You add, delete, modify elements within your database through PHP. Access cookies variables and set cookies. Using PHP, you can restrict users from accessing some pages of your website. It can encrypt data.

2.MySQL

MySQL is a fast, easy-to-use RDBMS being used for many small and big businesses. MySQL is developed, marketed and supported by MySQL AB, which is a Swedish company. MySQL is becoming so popular because of many good reasons:

- MySQL is released under an open-source license. So, you have nothing to pay to use it.
- MySQL is a very powerful program in its own right. It handles a large subset of the functionality of the most expensive and powerful database packages
- MySQL works on many operating systems and with many languages including PHP, PERL, C, C++, JAVA, etc.
- MySQL works very quickly and works well even with large data sets. MySQL is very friendly to PHP, the most appreciated language for web development. MySQL is customizable.
- MySQL supports large databases, up to 50 million rows or more in a table. The default file size limit for a table is 4GB, but you can increase this (if your operating system can handle it) to a theoretical limit of 8 million terabytes (TB).

2.5.2 HARDWARE SPECIFICATIONS

- ❖ Processor: Intel Core i3 or equivalent
- ❖ Memory: Minimum of 4GB RAM
- ❖ Storage: 250 GB or above

2.5.3 SOFTWARE SPECIFICATIONS

- ❖ Front-End: HTML5 with CSS3 and JavaScript
- ❖ Back-End: PHP for server-side scripting
- ❖ Database: MySQL 5.7 with phpMyAdmin 4.8 server interface
- ❖ Operating System: Windows 8 or higher
- ❖ Email Service: SMTP protocol or third-party services (SendGrid, AWS SES)
- ❖ Web Browser: Google Chrome (Version 67 or later)
- ❖ Server: XAMPP or WAMP Server 3.1
- ❖ Other Tools: PDF Reader and Antivirus software

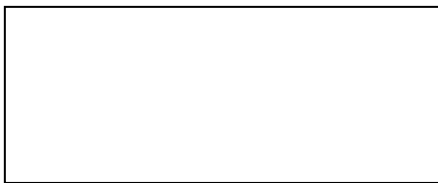
3. SYSTEM DESIGN AND DEVELOPMENT

3.1 DATAFLOW DIAGRAM

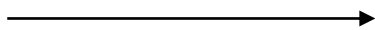
A Data Flow Diagram (DFD) is a visual representation that illustrates how data flows through a system. It shows the processes, data stores, external entities, and data interactions within the system, providing a clear and structured view of how information is processed and handled.

Here it shows data flow in and between Client, Trainer, Admin as well as DFD models how the various components of the site interact with each other.

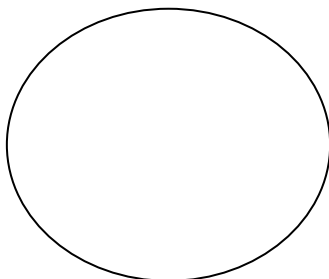
Basic data flow diagram symbols are:



A "Rectangle" defines a source (originator) or destination of data.



An "Arrow" identifies the data flow. It is a pipeline through which information flow

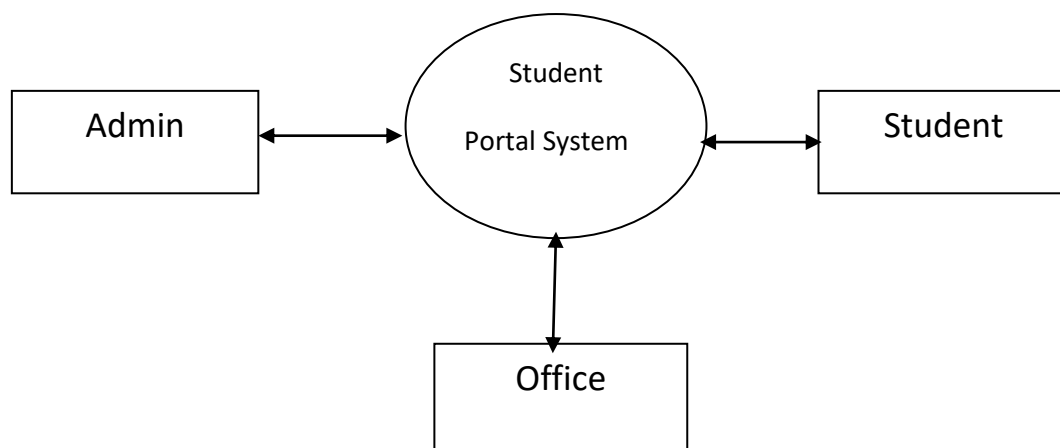


A "Circle" represents a process that transforms incoming data flow(s).



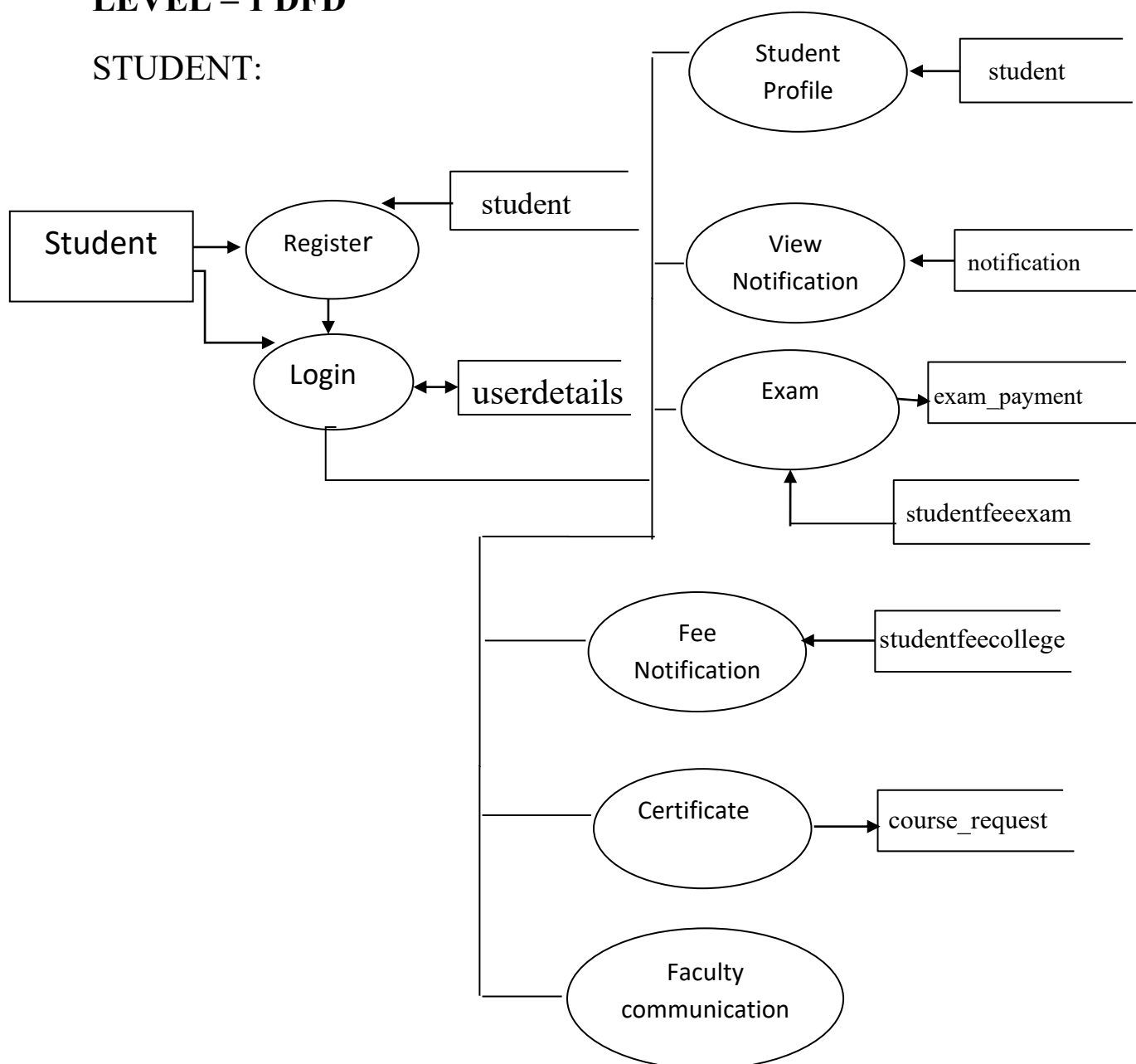
An "Open Rectangle" is a data store.

LEVEL – 0 DFD



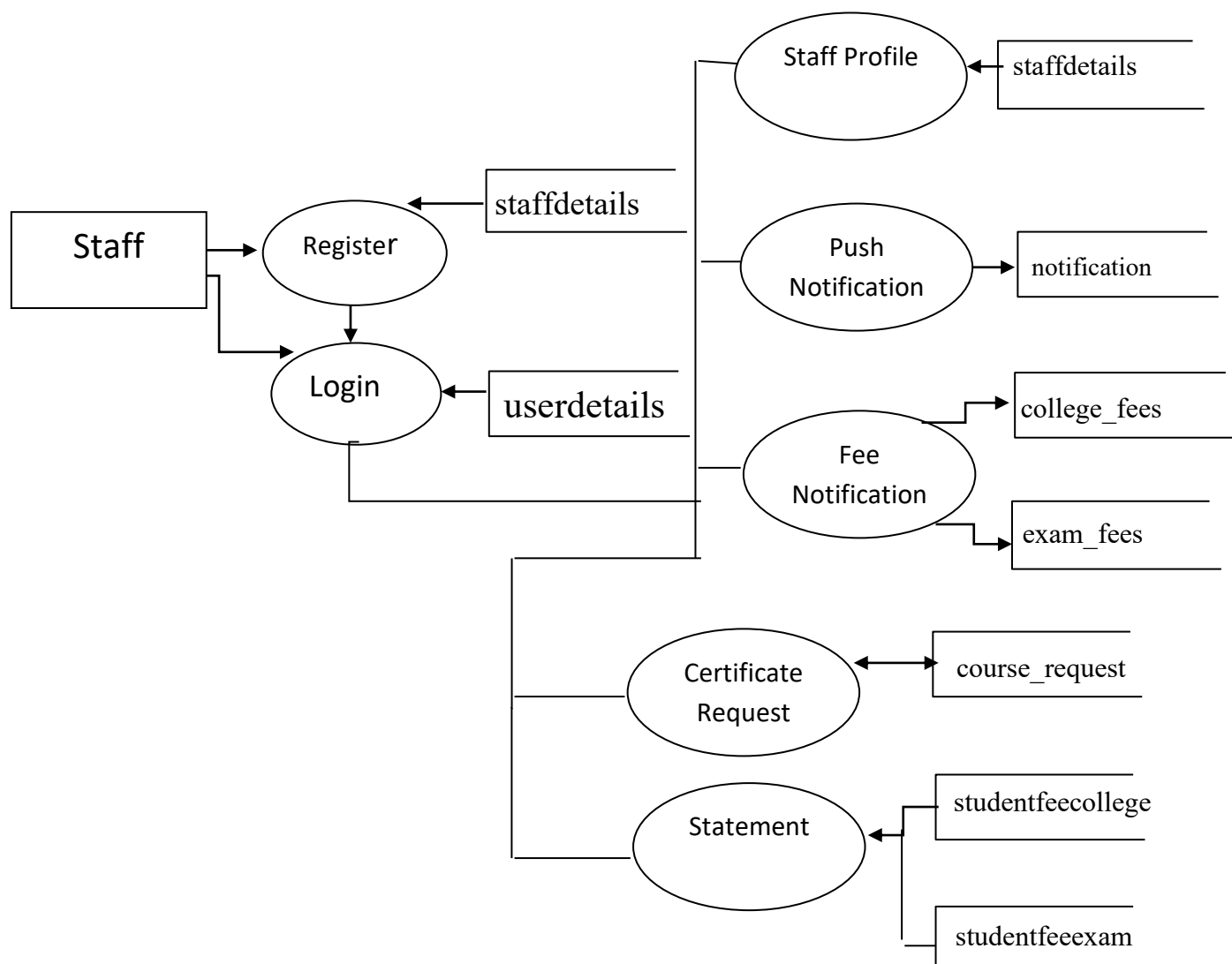
LEVEL – 1 DFD

STUDENT:



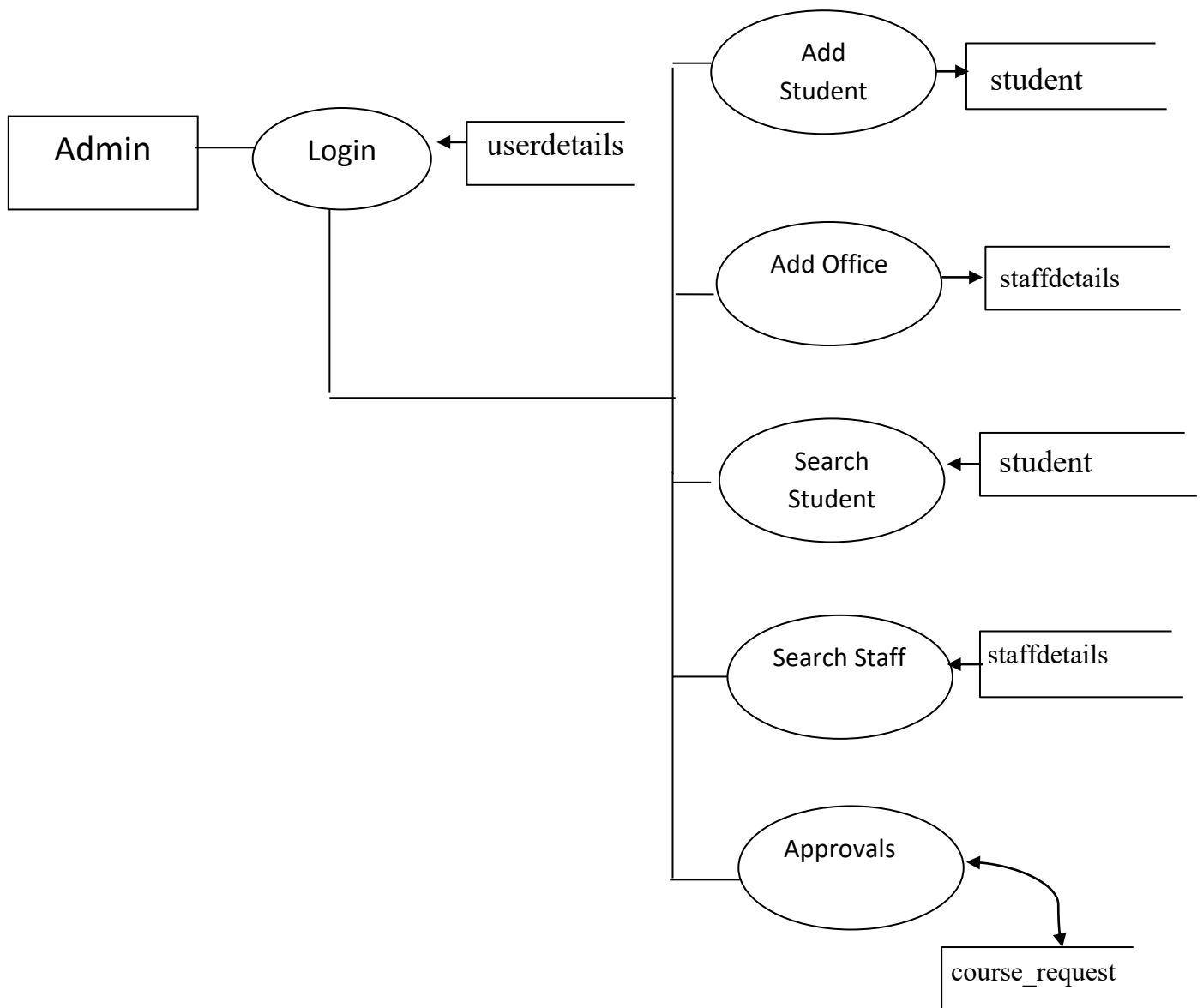
LEVEL – 1 DFD

OFFICE:



LEVEL – 1 DFD

ADMIN:



3.2 DATABASE DESIGN

The data in the system is stored and retrieved from the database. Designing the database is a part of the system design. Data elements and data structures to be stored have been identified at analysis stage. They are structured and put together to design the data storage and retrieval system. The general theme behind the database is to handle the information as an integrated whole. When designing a database, we must make decisions regarding how best to make some systems in the real world and model it as database.

The goal of database design is to generate a set of relations that allows storing information easily. The database is designed in relational model in which the data are organized into entries and relation between them.

In our data flow diagrams, we have given names to data flows, process and data stores. Although the names are descriptive of the data, they do not give details. So the following DFD gives the details of the fields used. A data dictionary has many advantages in improving analyst of user communication by establishing consistent definitions of terms elements and procedures.

TABLE DESIGN

Table Name: userdetails

Fields Name	Data Type	Null	Description
id	varchar(10)	No	Primary key for all users
username	varchar(23)	No	Login username
password	varchar(20)	No	Login username
role	enum('admin','student','staff')	No	Defines user role
profile_pic	varchar(100)	Yes	Profile picture path

Table Name: student

Fields Name	Data Type	Null	Description
admissionNo	varchar(10)	No	Primary key – admission number
fullname	varchar(20)	No	Fullname
regNo	bigint(20)	No	University Registration number
department	varchar(23)	No	Department name
semester	int(11)	Yes	Current semester
PhnNo	bigint(20)	Yes	Phone number
email	varchar(50)	Yes	Email address
ParentName	varchar(20)	Yes	Parent's name

ParentPho	bigint(20)	Yes	Parent's phone number
address	varchar(40)	Yes	Address
Dob	date	Yes	Date of Birth
category	varchar(20)	No	Category(General/SC/ST)
Gender	varchar(20)	No	Gender
bloodgroup	varchar(20)	No	Blood group
joiningdate	date	Yes	Admission date
year	int(11)	No	Academic year
user_id	varchar(10)	Yes	FK- userdetails.id

Table Name: staffdetails

Fields Name	Data Type	Null	Description
staffid	varchar(10)	No	Primary key – staff Id
staffname	varchar(20)	No	Staff name
sstaffrole	varchar(20)	Yes	Role/designation
department	varchar(25)	Yes	Department
address	varchar(30)	Yes	Address
PhnNo	bigint(20)	Yes	Phone number
email	varchar(50)	Yes	Email id
profile_pic	varchar(100)	No	Profile image
user_id	varchar(10)	Yes	FK- userdetails.id

Table Name: college_fee

Fields Name	Data Type	Null	Description
fee_id	int(11)	No	Primary key – fee template Id
fee_title	varchar(100)	Yes	Fee title
department	varchar(30)	Yes	Applicable department
semester	int(11)	Yes	Applicable semester
year	int(11)	Yes	Applicable year
category	varchar(50)	Yes	Category
amount	int(11)	Yes	Fee amount
last_date	date	Yes	Payment deadline
active_flag	tinyint(1)	Yes	Active status
fee_structure_path	varchar(255)	Yes	File path for structure

Table Name: studentfeecollege

Fields Name	Data Type	Null	Description
id	int(11)	No	Primary key
student_id	varchar(10)	No	FK – student.admissionNo
feeId	int(11)	No	FK – college_fee.fee_id
FeeTitle	varchar(30)	No	Fee title
department	varchar(30)	Yes	Department
semester	int(11)	Yes	Semester
year	int(11)	Yes	Year

category	varchar(20)	Yes	Category
amount	int(11)	Yes	Amount
lastdate	date	Yes	Last payment date
status	enum('pending', 'paid')	Yes	Payment status
upi_id	varchar(50)	Yes	UPI ID

Table Name: exam_fees

Fields Name	Data Type	Null	Description
id	int(11)	No	Primary key
exam_id	varchar(50)	Yes	Exam Id (unique)
exam_name	varchar(150)	Yes	Exam name
department	varchar(30)	Yes	Department
semester	int(11)	Yes	Semester
category	varchar(100)	Yes	Category
amount	decimal(10,2)	Yes	Exam Fee amount
due_date	date	Yes	Due date
fee_structure_path	varchar(255)	Yes	File path for structure
active_flag	tinyint(1)	Yes	Active status

Table Name: studentfeeexam

Fields Name	Data Type	Null	Description
id	int(11)	No	Primary key
studentid	varchar(10)	No	FK- student.admissionNo
studentRoll	bigint(20)	No	Roll number
department	varchar(30)	Yes	Department
category	varchar(30)	Yes	Category
semester	int(11)	Yes	Semester
examid	varchar(20)	Yes	FK - exam_fees.exam_id
examname	text	Yes	Exam name
amount	decimal(11,0)	Yes	Amount
lastdate	date	Yes	Due date
status	enum('pending','paid')	Yes	Payment status
upi_id	varchar(50)	No	UPI ID

Table Name: exam_payments

Fields Name	Data Type	Null	Description
id	int(11)	No	Primary key
student id	varchar(10)	No	FK- student.admissionNo
rollno	int(11)	No	Roll no
student_name	varchar(100)	No	Student name
department	varchar(100)	No	Department

semester	varchar(20)	No	Semester
exam_code	varchar(50)	No	FK- exam_fees.exam_id
exam_name	varchar(100)	No	Exam name
amount	decimal(10,2)	No	Paid amount
upi_id	varchar(100)	Yes	UPI ID
payment_date	datetime	Yes	Payment date
status	enum('paid','pending')	Yes	Payment status

Table Name: course_requests

Fields Name	Data Type	Null	Description
id	int(11)	No	Primary key
admissionNo	varchar(10)	No	FK- student.admissionNo
name	varchar(100)	No	Student name
department	varchar(100)	No	Department
course	varchar(100)	No	Course name
year	int(11)	No	Year
message	text	Yes	Message
signature_path	varchar(255)	Yes	Signature file
doc1_path	varchar(255)	Yes	Document 1
doc2_path	varchar(255)	Yes	Document 2
request_date	date	Yes	Request date

stage	enum('Pending', 'Verified','Approved')	Yes	Verification stage
status	enum('Pending', 'Rejected','Approved')	Yes	Approval status
rejection_message	text	Yes	Rejection message
certificatepath	varchar(255)	Yes	Certificate path

Table Name: notifications

Fields Name	Data Type	Size	Description
id	int(11)	No	Primary key
title	varchar(255)	No	Notification title
department	varchar(100)	No	Department
message	text	No	Message content
created_at	date	Yes	Created date

3.3 INPUT AND OUTPUT SCREENS

INPUT DESIGN

Input design is a process of converting a user-oriented description of inputs to a computer-based program-oriented specification. The main objective is to create an input layout that is easy to follow and avoid entering data incorrectly.

The valid data in turn is stored as operational data in the database. The goal of input design is to make input data entry as easy and error free. Input screen takes care to filter the invalid data from becoming an operational data at data entry phase.

Menu Driven

The system is menu driven. This simplifies computer data access or data entry. Instead of remembering, what to enter, the user chooses from a list of options by positioning the cursor appropriately. This is adopted for menus with a few options to select from. The menu limits a user choice of suspense but reduces the chances for errors in data entry.

Easy data input

Data entry screens have been designed in a manner much similar to old systems. Each form has controls for insertion, updating and exit. Appropriate messages are provided in the message area, which prompts the user in entering the right data. Erroneous data inputs are checked while user inputs.

Data Validation

The input data is validated for minimize errors in data entry. For certain data specific codes have been given and validation is done which enable user to enter the required data or correct them if they have entered wrong codes. Similarly, validation can be done with respect to the data type. For example, entry of data fields, numeric fields etc.

Use Friendliness

User is never left in a state of confusion as to what is happening; instead, appropriate error and acknowledgement messages are sent. Error capturing is used to indicate the error codes and specific error messages.

Consistent Format

A fixed format is adopted for, displaying the title and messages. Every screen has buttons, which displays the operation that can be performed after data entry. They are normally done at the touch of a key or mouse.

Interactive dialogue

The system engages the user in an interactive dialogue. The system can extract missing or omitted information from the user by directing the user through appropriate messages, which are displayed.

Input design is the process of converting user-oriented inputs to computer-based format. It also includes determining the record media, method of input, speed of capture, and entry into the system.

OUTPUT DESIGN

The normal procedure is to design the outputs in detail first and then to work back to the inputs. The outputs can be in the form of operational documents, lengthy reports. The input records must be validated, edited, organized and accepted by the system before being processed to produce the outputs.

The next stage is to determine the most appropriate medium for output. Consideration will be:

- Suitability of the device to the application
- The need for hard copy
- The response time required
- The software/hardware available and the cost

1.Home Page



Fig.1

2.Registration

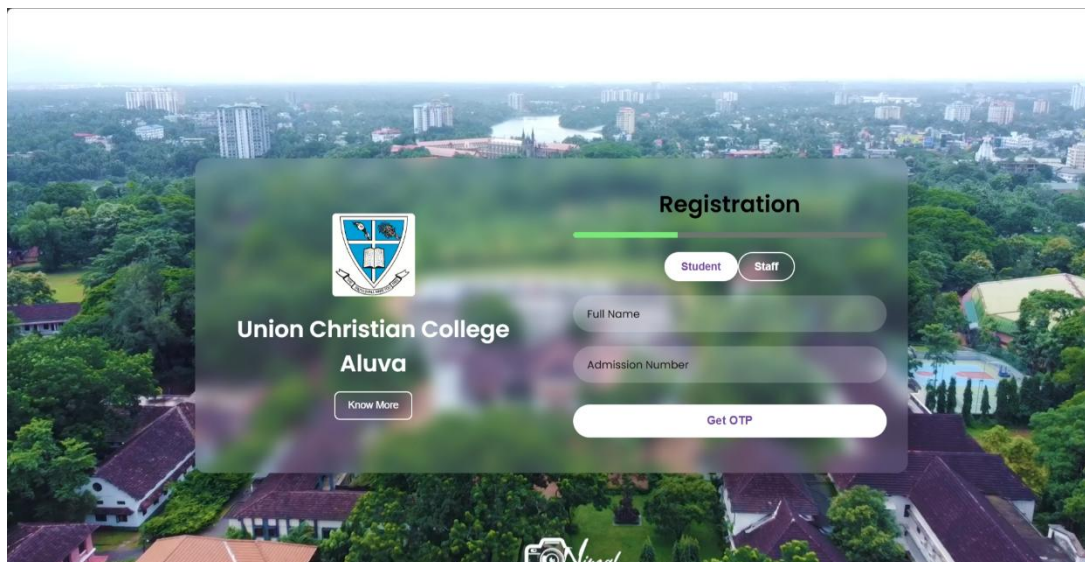


Fig.2

3.Successful Registration

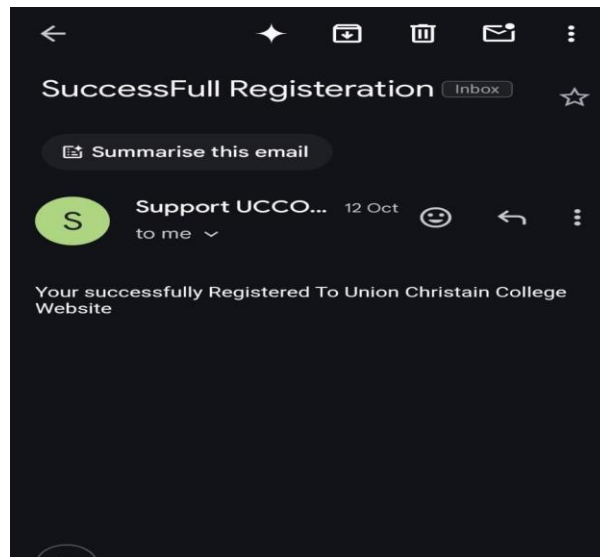


Fig.2.1

4.Login

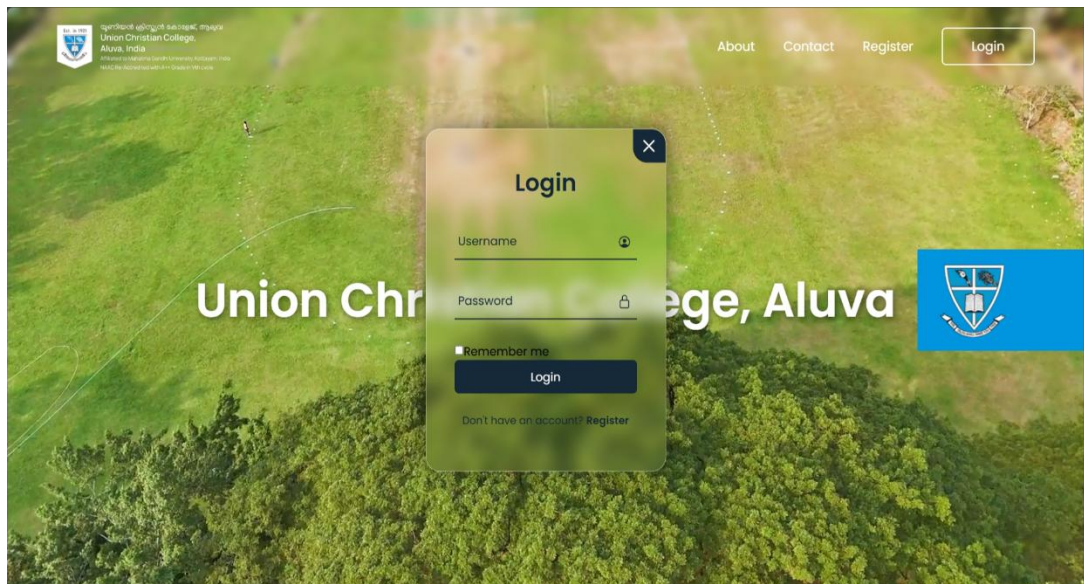
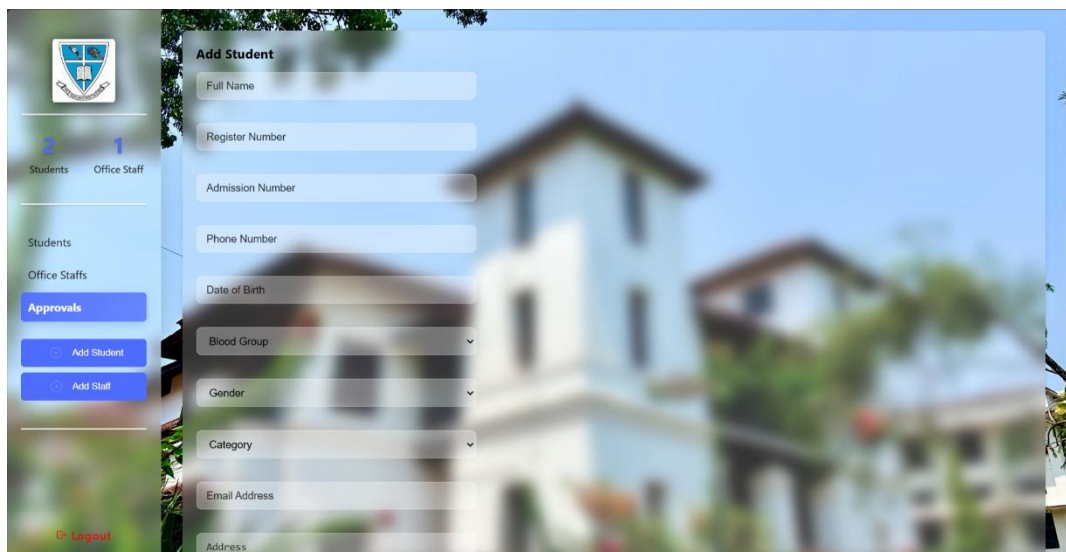


Fig.3

ADMIN PAGE

1.Add Student



Add Student

Full Name

Register Number

Admission Number

Phone Number

Date of Birth

Blood Group

Gender

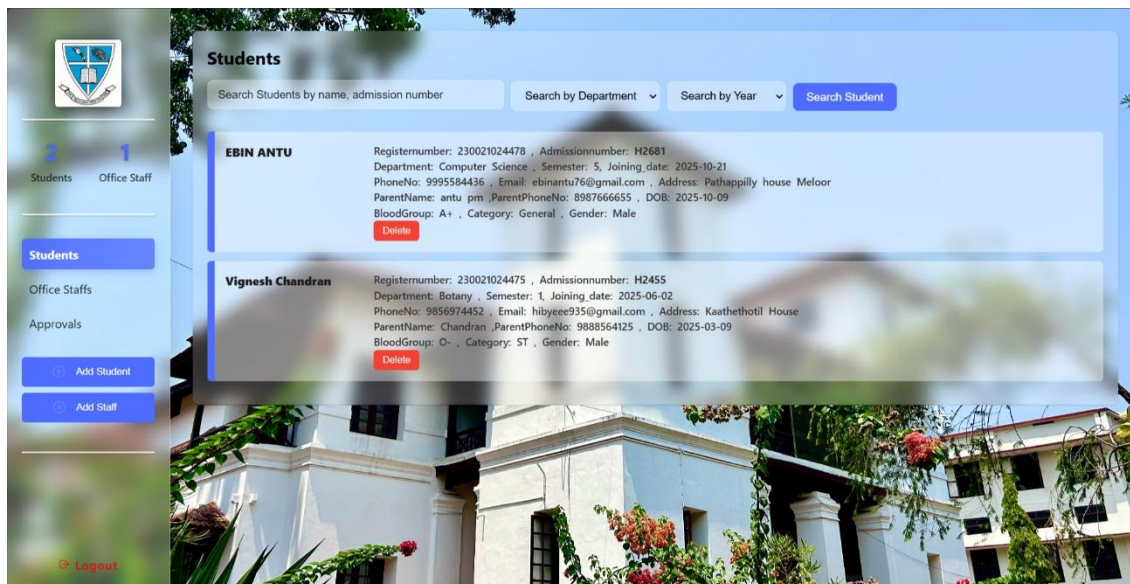
Category

Email Address

Address

Fig.1

2.Students



Students

Search Students by name, admission number Search by Department Search by Year [Search Student](#)

EBIN ANTU Registernumber: 230021024478 , Admissionnumber: H2681
 Department: Computer Science , Semester: 5, Joining date: 2025-10-21
 PhoneNo: 9995584436 , Email: ebinantu76@gmail.com , Address: Pathappilly house Meloor
 ParentName: antu pm , ParentPhoneNo: 8987666655 , DOB: 2025-10-09
 BloodGroup: A+ , Category: General , Gender: Male
[Delete](#)

Vignesh Chandran Registernumber: 230021024475 , Admissionnumber: H2455
 Department: Botany , Semester: 1, Joining date: 2025-06-02
 PhoneNo: 9856974452 , Email: hilyee935@gmail.com , Address: Kaathetholil House
 ParentName: Chandran , ParentPhoneNo: 9888564125 , DOB: 2025-03-09
 BloodGroup: O- , Category: ST , Gender: Male
[Delete](#)

Fig.1.1

3.Add Office Staff

Add Office Staff

Staff Name

Staff ID

Staff Role

Department

Address

Phone Number

Email Address

Add Office Staff

Fig.2

4.Office Staff

Office

2 Search Staff

Dhony ID: 22222222, Role: Manager
Dept: Finance and Accounts, Phone: 7306389589
Email: dhonythomas937@gmail.com
Address: Kallloor House
Delete

Fig.2.1

5.Approvals

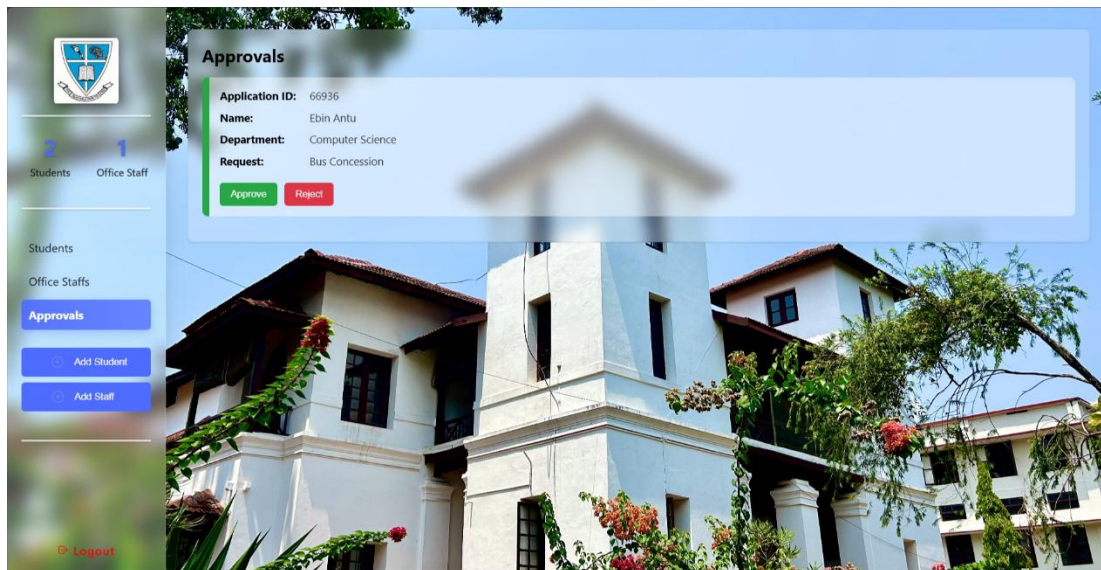


Fig.3

6.Approve Request and Upload Certificate

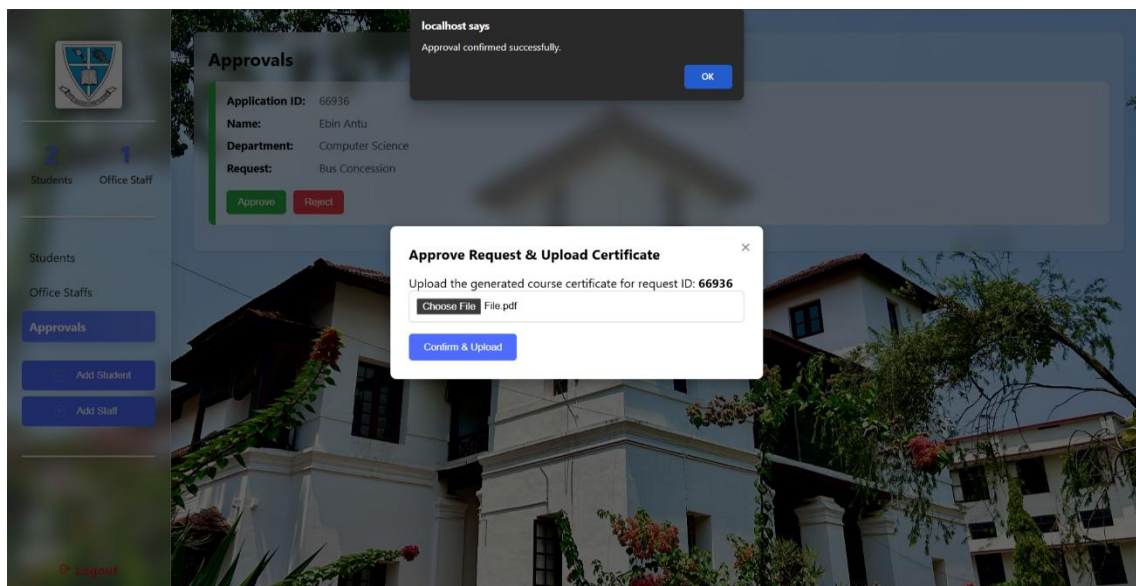


Fig.3.1

7.Reject Request

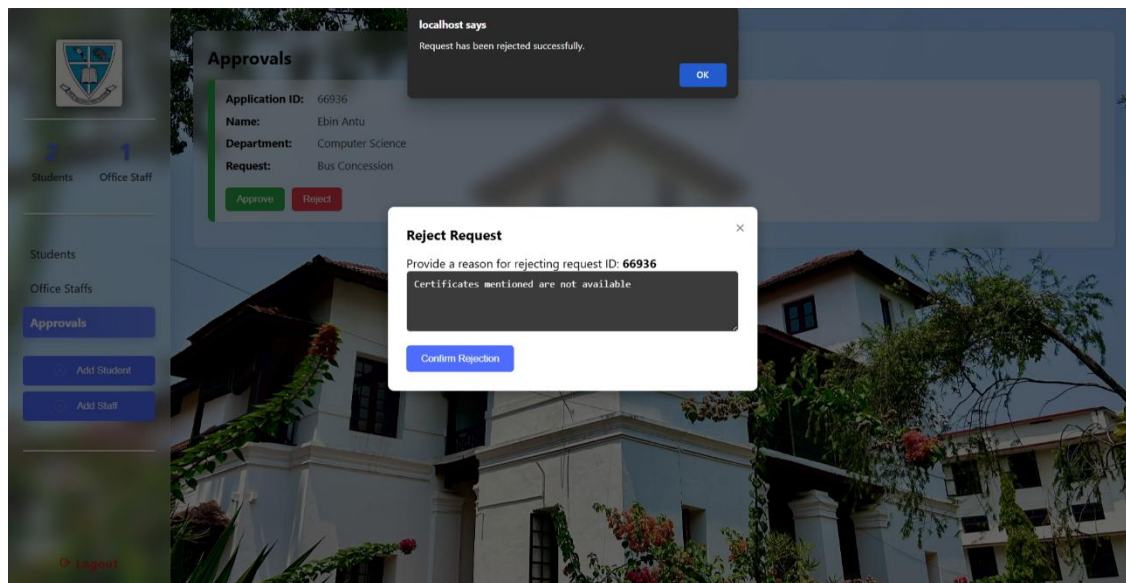


Fig.3.2

OFFICE PAGE

1.Profile Details

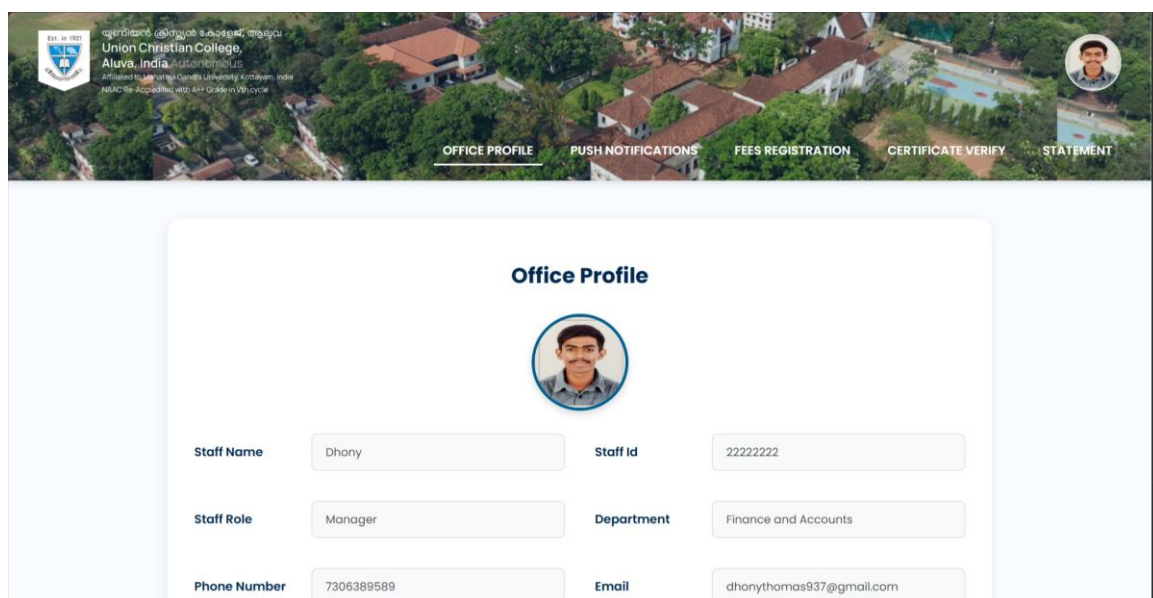
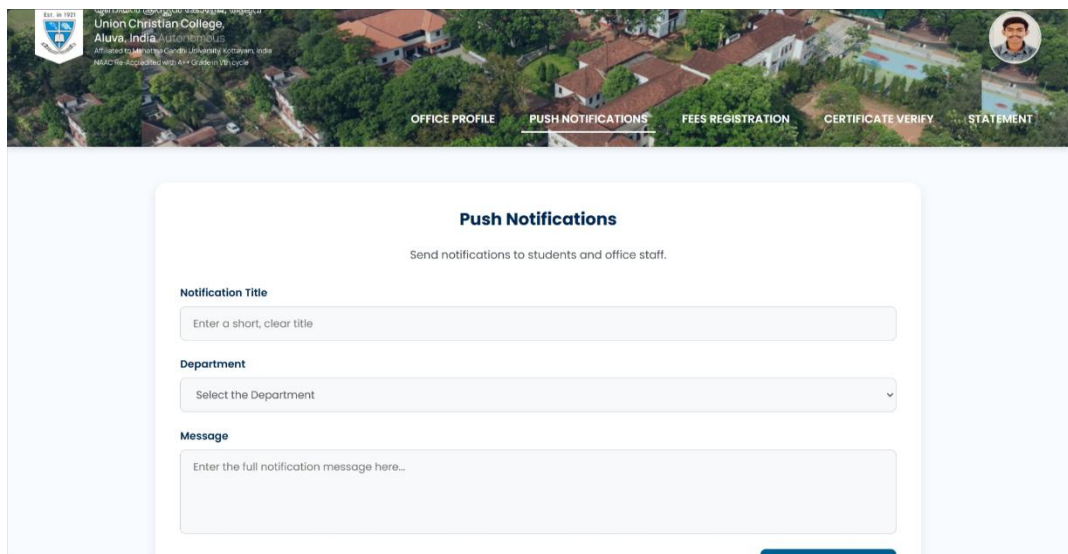


Fig.1

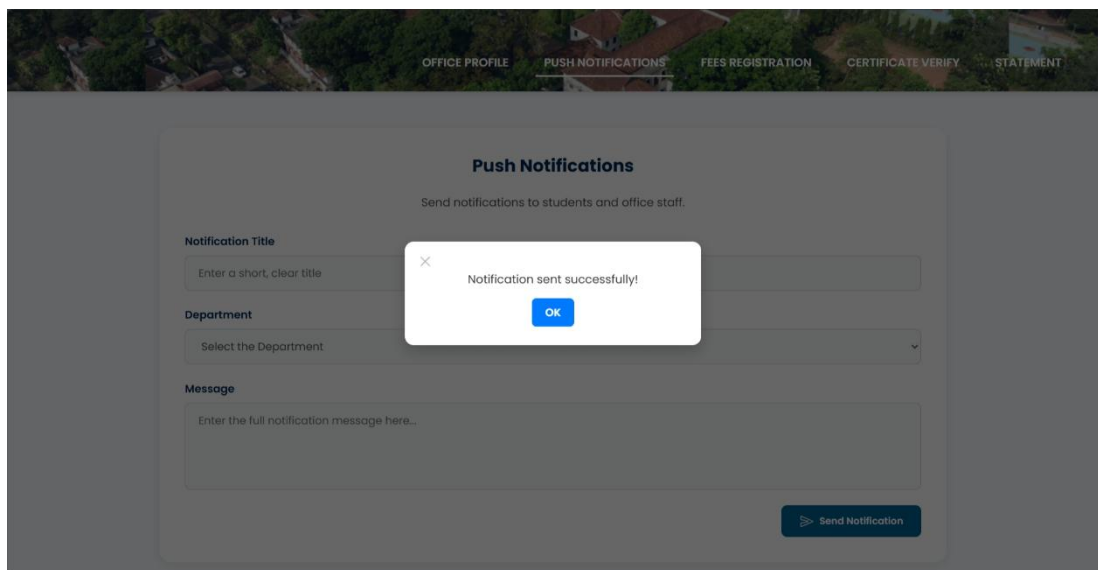
2. Push Notifications



The screenshot shows the 'Push Notifications' page of the Union Christian College portal. The header includes the college logo, name, and address, along with navigation links: OFFICE PROFILE, PUSH NOTIFICATIONS (active), FEES REGISTRATION, CERTIFICATE VERIFY, and STATEMENT. A user profile picture is visible in the top right. The main form area is titled 'Push Notifications' with the subtitle 'Send notifications to students and office staff.' It contains three input fields: 'Notification Title' with a placeholder 'Enter a short, clear title', 'Department' with a dropdown menu 'Select the Department', and 'Message' with a placeholder 'Enter the full notification message here...'. A blue 'Send Notification' button is at the bottom right of the form.

Fig.2

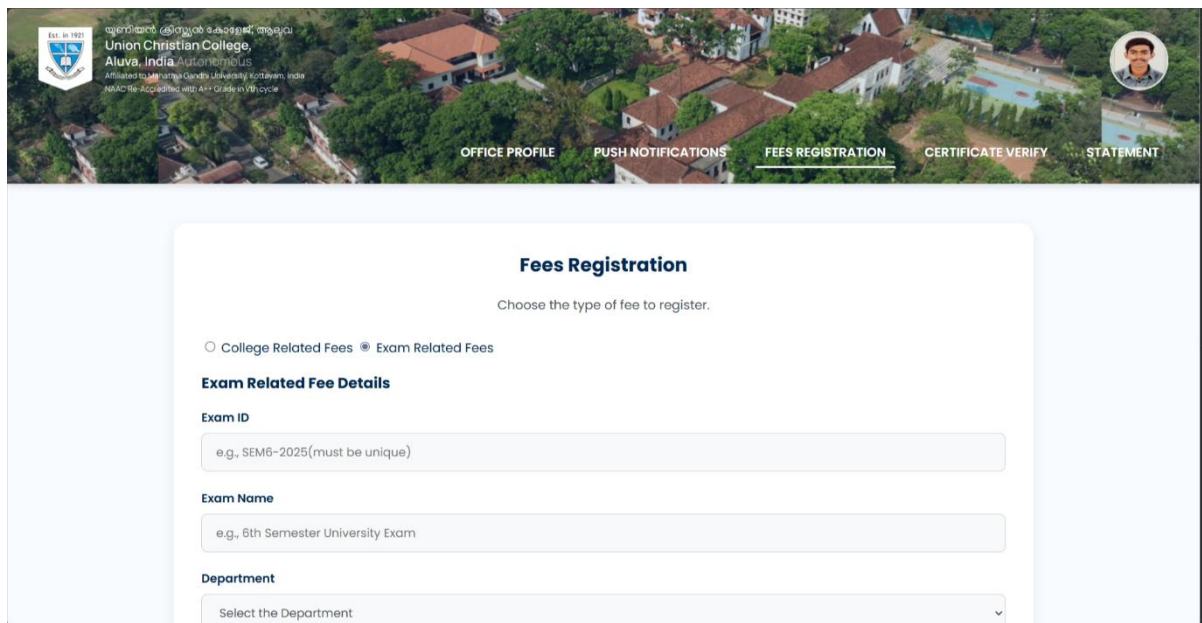
3. Notifications sent successfully



This screenshot shows the same 'Push Notifications' form as Fig.2, but with a success message overlay. The overlay is a white box with a close button (X) in the top left, the text 'Notification sent successfully!', and an 'OK' button. The background form is dimmed, showing the same fields and the 'Send Notification' button.

Fig.2.1

4. Fee Registration (Exam related)



Union Christian College,
Aluva, India Autonomous
Affiliated to Mar Thoma Syrian University, Kottayam, India
AACSB Accredited with A++ Grade in Vth Cycle

OFFICE PROFILE PUSH NOTIFICATIONS **FEES REGISTRATION** CERTIFICATE VERIFY STATEMENT

Fees Registration

Choose the type of fee to register.

☐ College Related Fees ☒ Exam Related Fees

Exam Related Fee Details

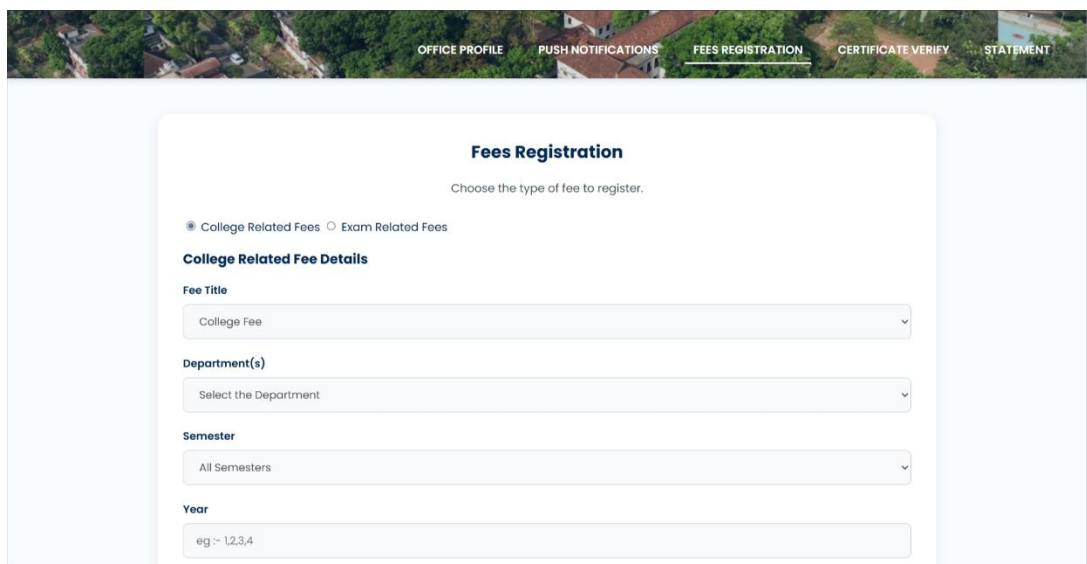
Exam ID
e.g., SEM6-2025 (must be unique)

Exam Name
e.g., 8th Semester University Exam

Department
Select the Department

Fig.3

5. College related fee



OFFICE PROFILE PUSH NOTIFICATIONS **FEES REGISTRATION** CERTIFICATE VERIFY STATEMENT

Fees Registration

Choose the type of fee to register.

☒ College Related Fees ☐ Exam Related Fees

College Related Fee Details

Fee Title
College Fee

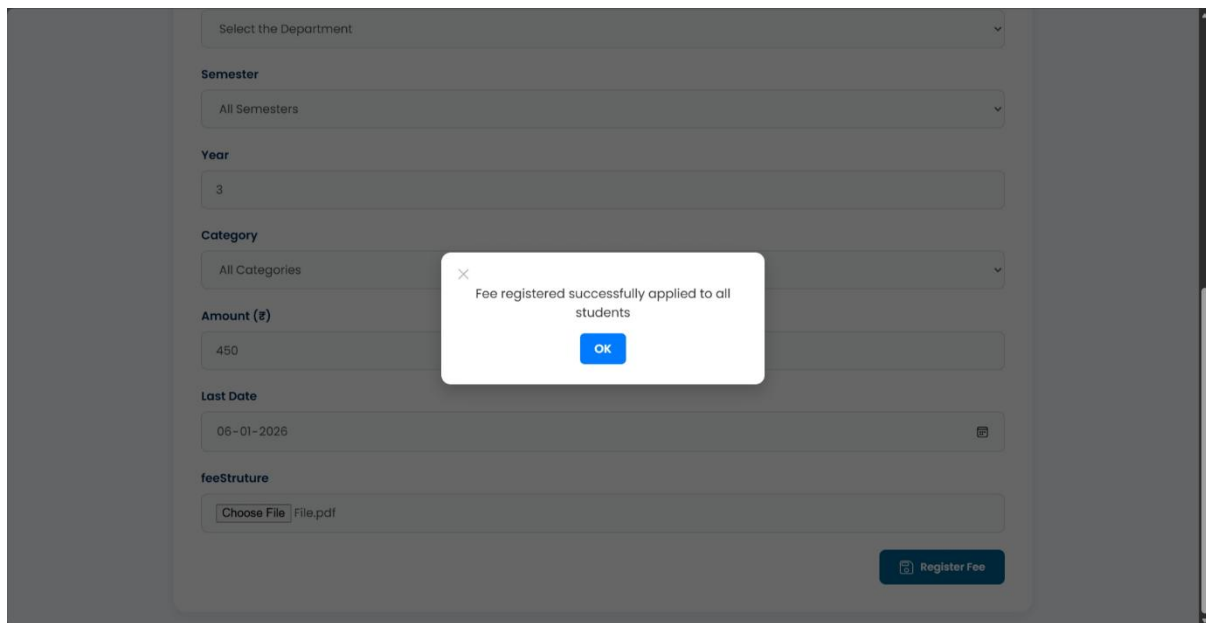
Department(s)
Select the Department

Semester
All Semesters

Year
eg :- 1,2,3,4

Fig.3.1

6. Fee Registered successfully

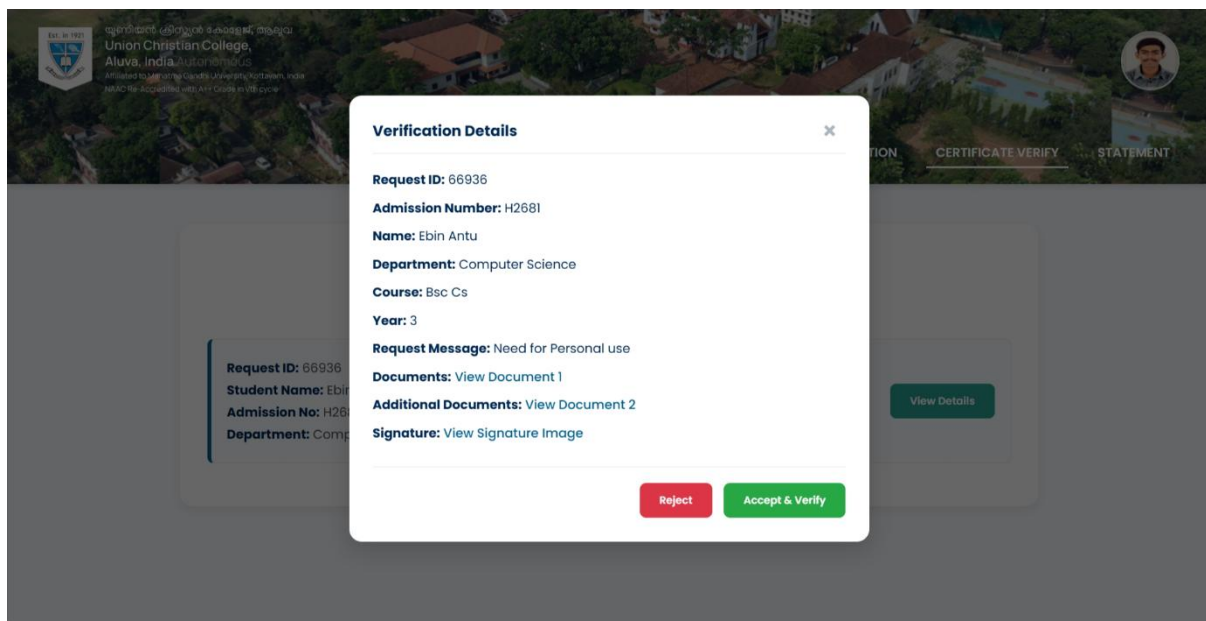


The screenshot shows a web form for fee registration. A modal dialog box is centered on the screen, displaying the message: "Fee registered successfully applied to all students" with an "OK" button. The background form is dimmed and includes the following fields:

- Select the Department (dropdown)
- Semester (dropdown, currently showing "All Semesters")
- Year (input field, currently showing "3")
- Category (dropdown, currently showing "All Categories")
- Amount (₹) (input field, currently showing "450")
- Last Date (input field, currently showing "06-01-2026")
- feeStruture (input field with "Choose File" and "File.pdf" options)
- A "Register Fee" button at the bottom right.

Fig.3.2

7. Verification Details



The screenshot shows a "Verification Details" modal dialog box. The background is a blurred image of a college campus. The modal contains the following information:

- Request ID:** 66936
- Admission Number:** H2681
- Name:** Ebin Antu
- Department:** Computer Science
- Course:** Bsc Cs
- Year:** 3
- Request Message:** Need for Personal use
- Documents:** [View Document 1](#)
- Additional Documents:** [View Document 2](#)
- Signature:** [View Signature Image](#)

At the bottom of the modal, there are two buttons: "Reject" (red) and "Accept & Verify" (green). In the background, a sidebar shows a list of requests with details like "Request ID: 66936", "Student Name: Ebin", "Admission No: H2681", and "Department: Comp".

Fig.4

8. Approval of Request

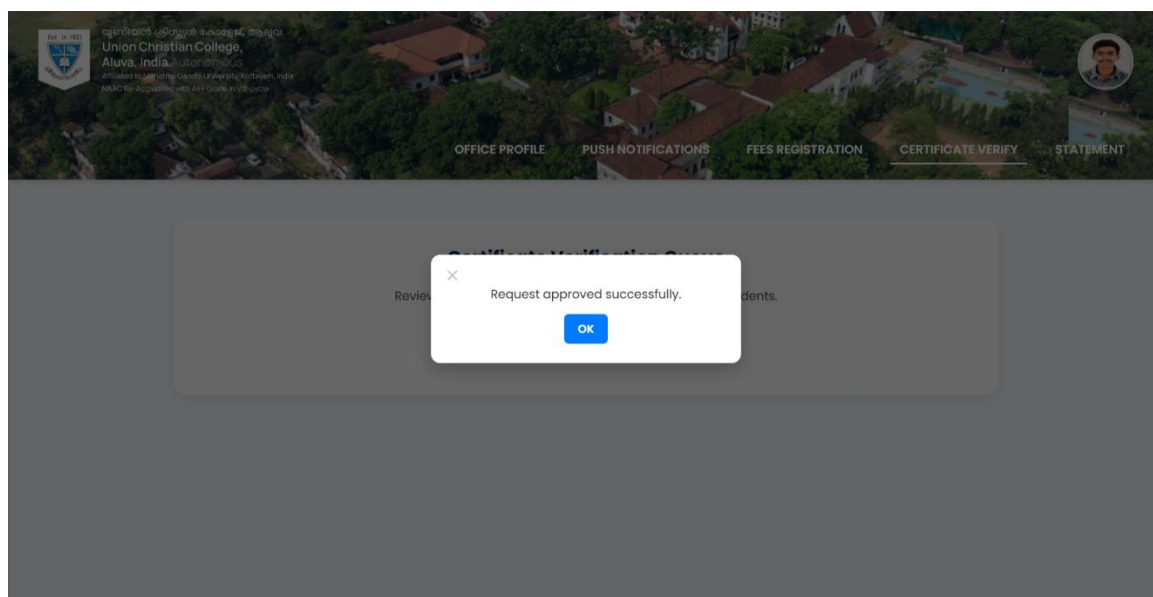


Fig.4.1

9. Verification Details

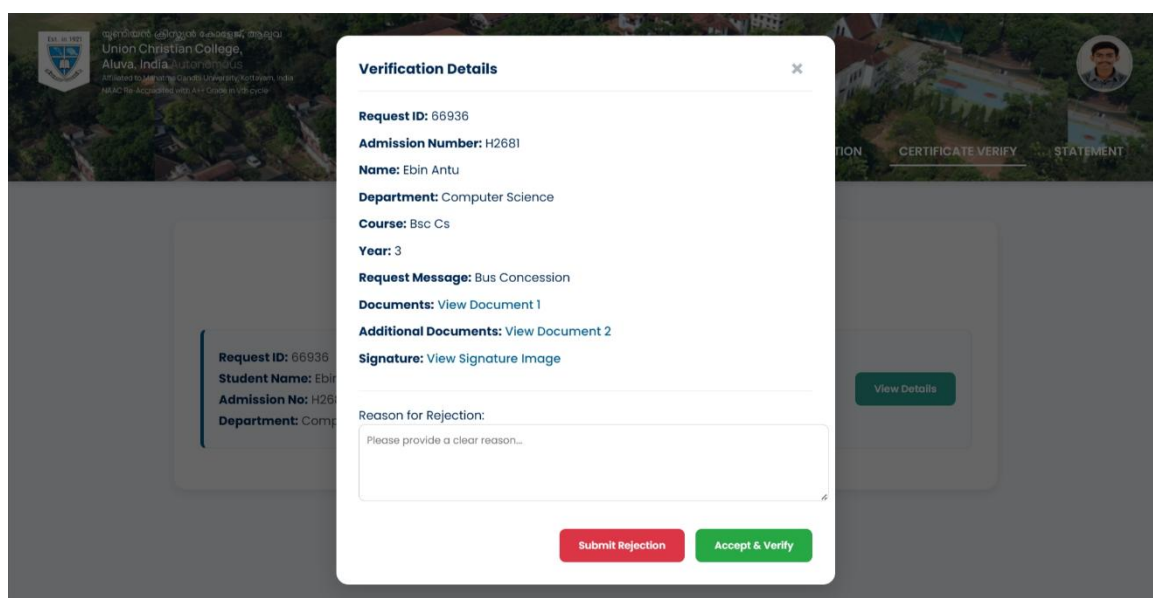


Fig.4.2

10.Rejection of Request

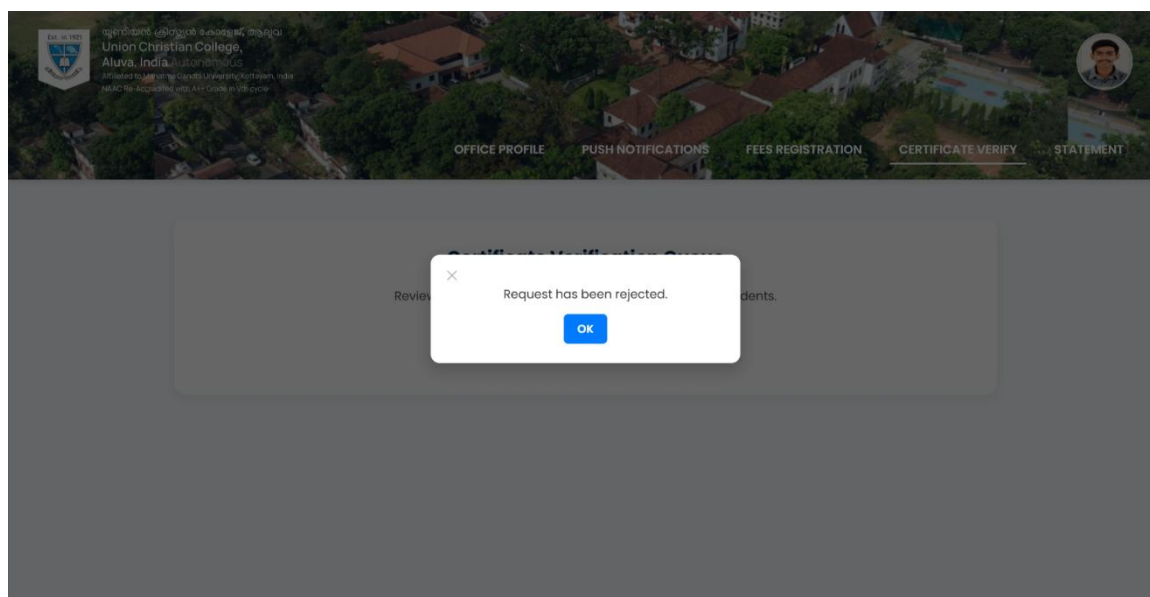


Fig.4.3

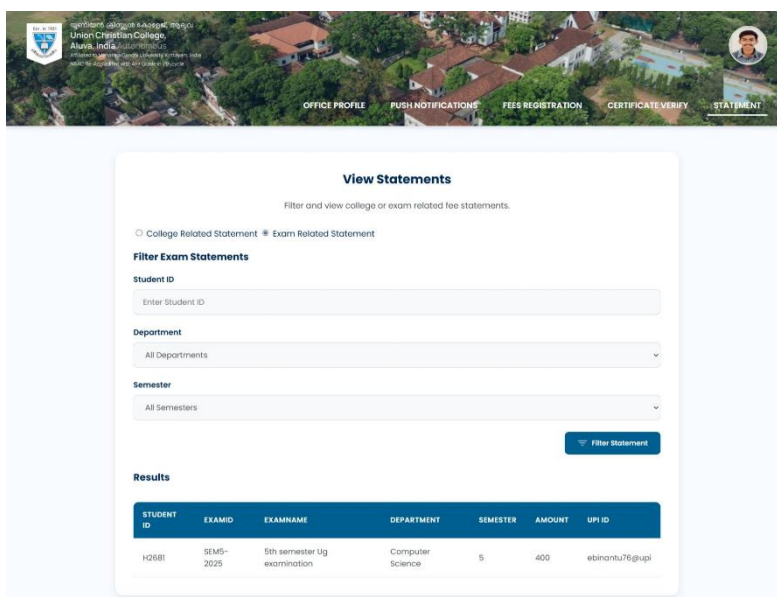
11.View Statements (College Related)

The screenshot shows the 'View Statements' form in the Union Christian College portal. The form is titled 'View Statements' and includes a subtitle 'Filter and view college or exam related fee statements.' Below the title, there are two radio buttons: 'College Related Statement' (selected) and 'Exam Related Statement'. Under the 'Filter College Statements' section, there are four dropdown menus: 'Fee Title' (set to 'All'), 'Department' (set to 'All Departments'), 'Semester' (set to 'All Semesters'), and 'Year' (set to 'e.g., 2025'). A 'Filter Statement' button is located at the bottom right of the filter section. Below the filter section, there is a 'Results' section with a table showing the filtered data.

STUDENT ID	FEE TITLE	DEPARTMENT	SEMESTER	YEAR	AMOUNT	UPI ID
H0681	College Fee	Computer Science	5	3	450	ebinarhu76@upi

Fig.5

12.Exam Related



View Statements
Filter and view college or exam related fee statements.

☐ College Related Statement ☒ Exam Related Statement

Filter Exam Statements

Student ID
Enter Student ID

Department
All Departments

Semester
All Semesters

[Filter Statement](#)

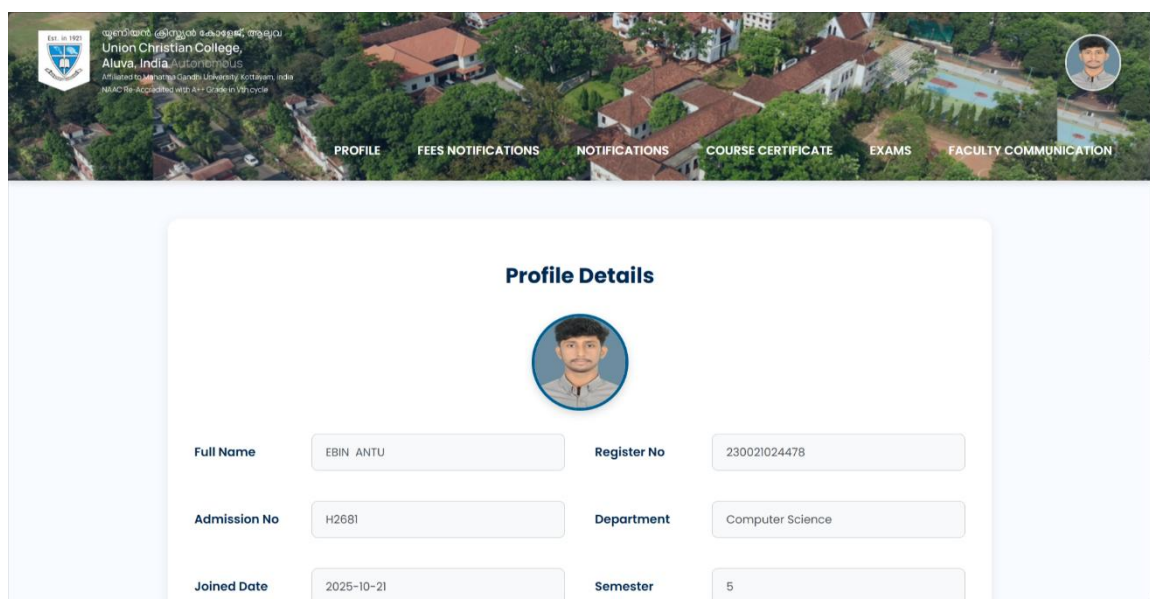
Results

STUDENT ID	EXAMID	EXAMNAME	DEPARTMENT	SEMESTER	AMOUNT	UPI ID
H2681	STMF-2025	5th semester UG examination	Computer Science	5	400	ebinantu76@upi

Fig.5.1

STUDENT PAGE

1.Profile Details



Profile Details



Full Name: EBIN ANTU Register No: 230021024478

Admission No: H2681 Department: Computer Science

Joined Date: 2025-10-21 Semester: 5

Fig.1

2.Fee Notifications

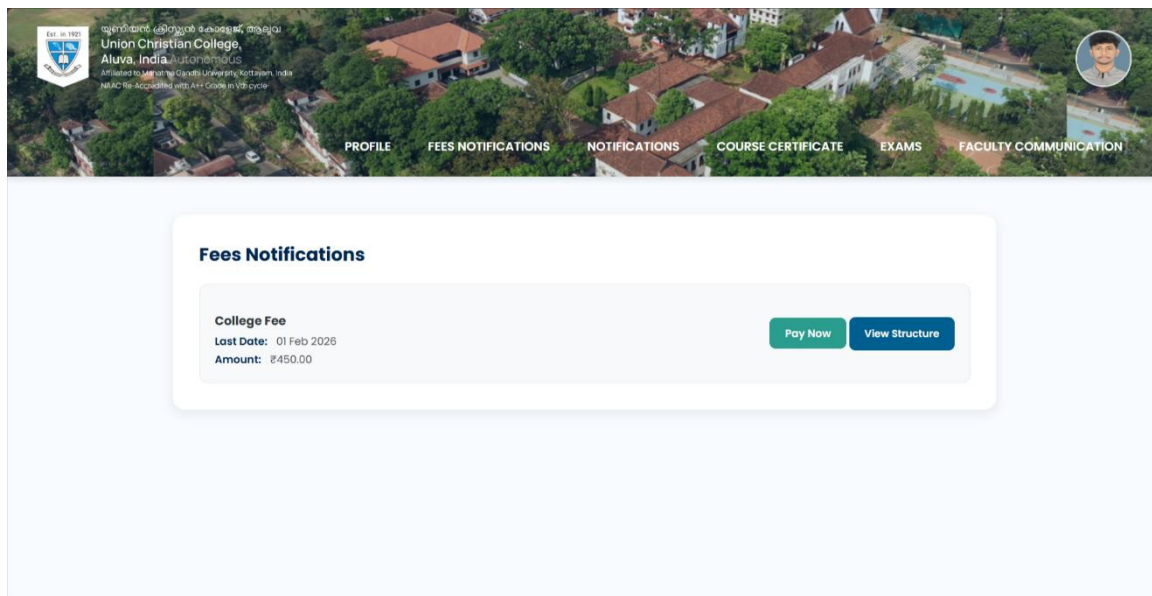


Fig.2

3.Payment Details Confirmation

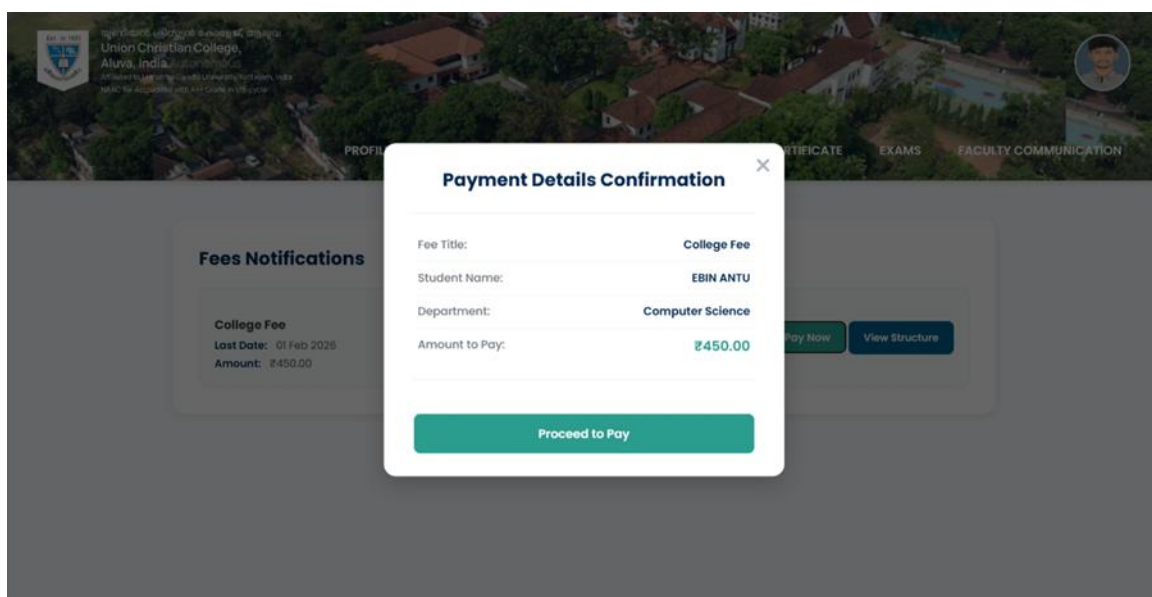


Fig.2.1

4.Payment QR Code

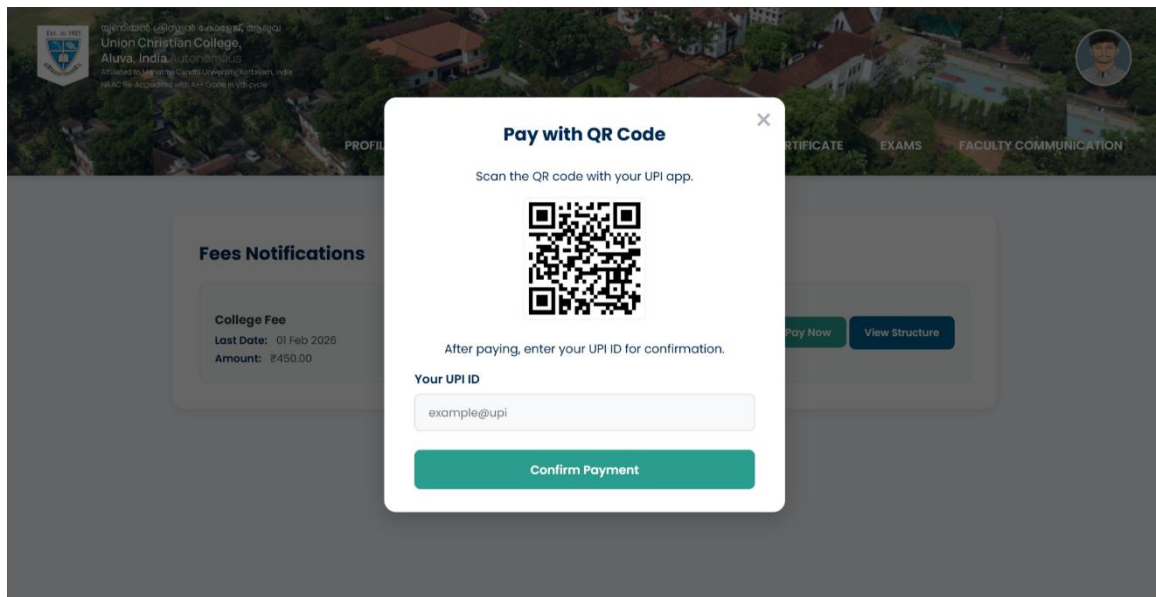


Fig.2.2

5.Payment Confirmed

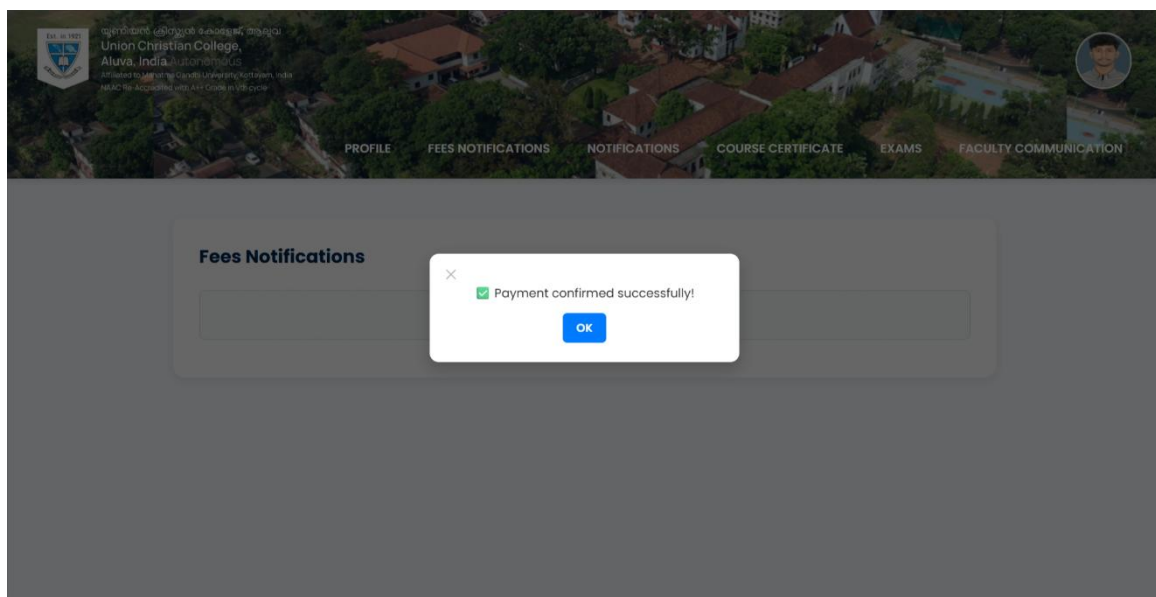


Fig.2.3

6. Notifications

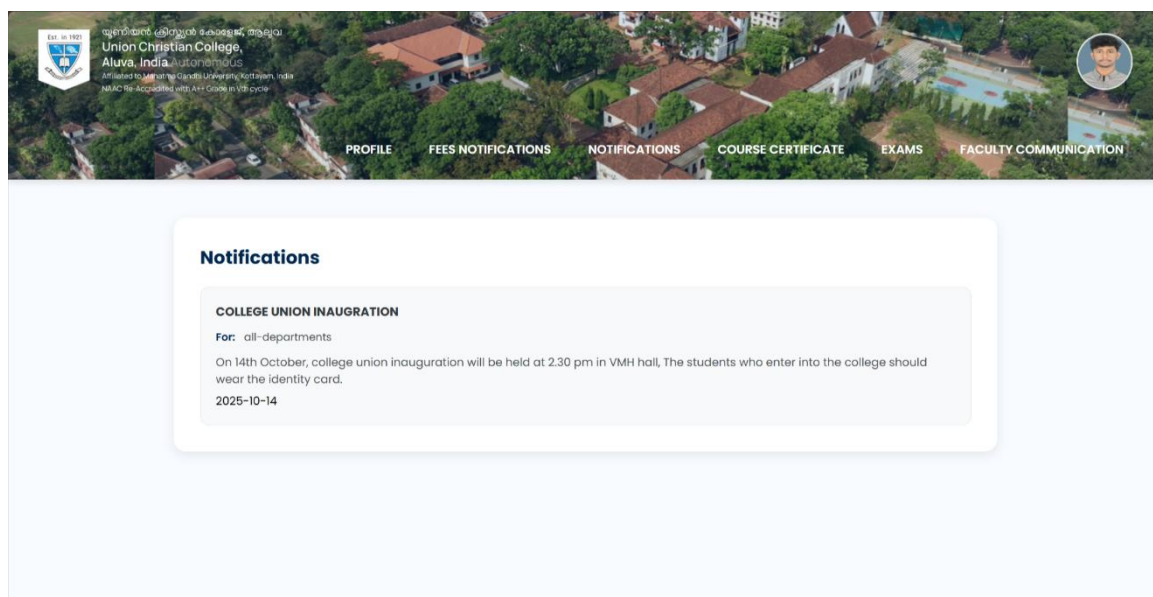


Fig.3

7. Course Certificate Request

The screenshot displays the 'Course Certificate Request' form. It begins with a note asking the user to fill out the form to request a course completion certificate. The form contains several input fields: 'Admission Number', 'Name', 'Department' (a dropdown menu), 'Course', and 'Year'. Below these is a 'Request Message' text area. At the bottom, there are two sections for 'Supporting Documents For Master verification', each with a 'Choose File' button and a 'No file chosen' status. A 'Submit Request' button is located at the bottom right of the form.

Fig.4

8.Application Code

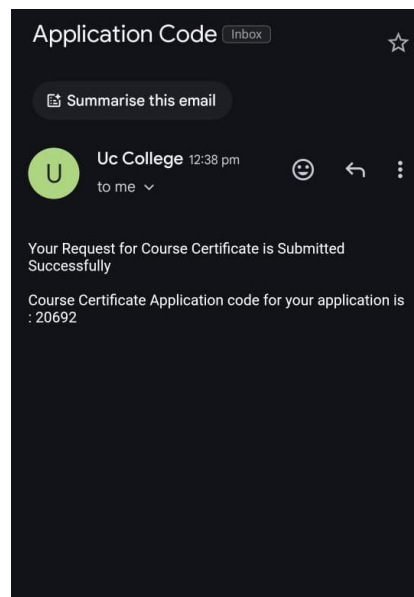


Fig.4.1

9.Application Status -Requested

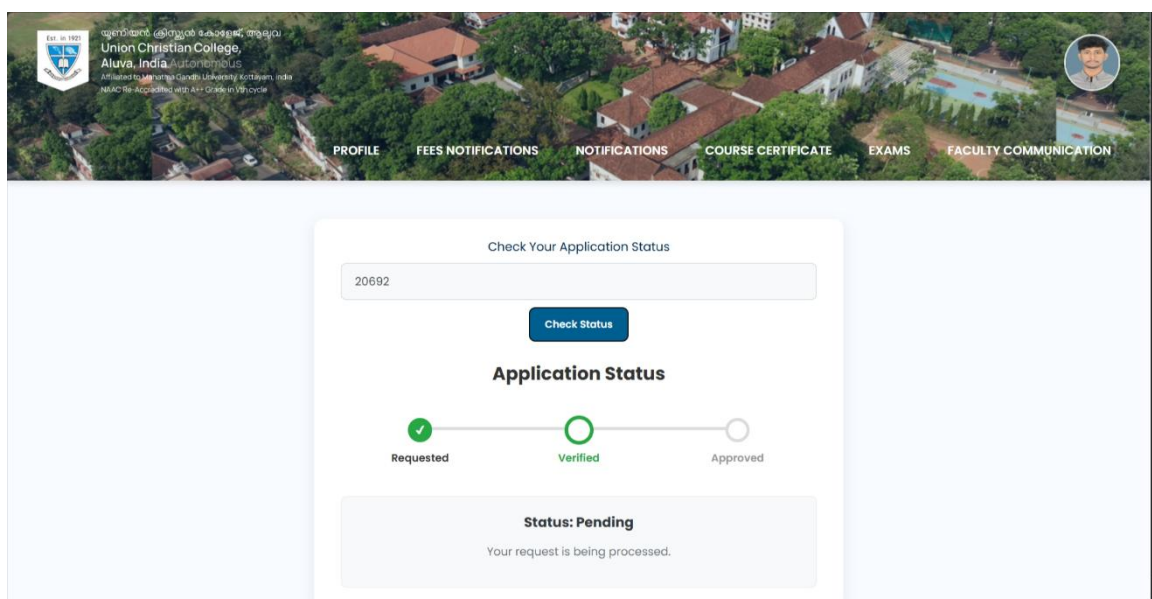
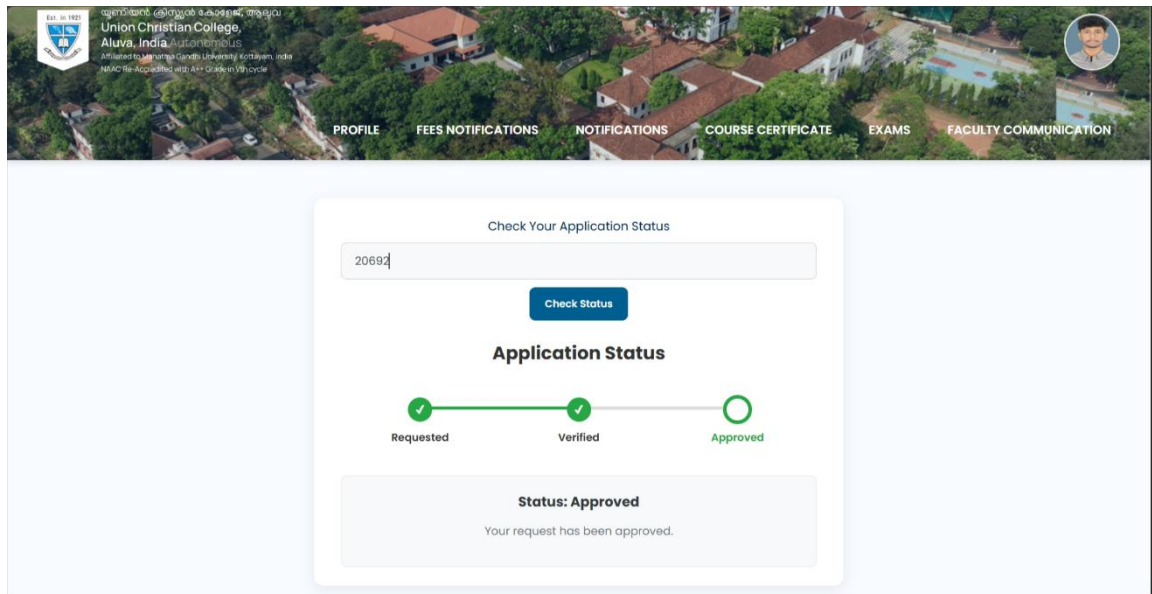


Fig.4.2

10.Verified



Union Christian College, Aluva, India. Autonomous. Affiliated to Mar Thoma Syrian University, Kottayam, India. NAAC Re-Accredited with A++ Grade in Vth Cycle.

PROFILE FEES NOTIFICATIONS NOTIFICATIONS COURSE CERTIFICATE EXAMS FACULTY COMMUNICATION

Check Your Application Status

20692

Check Status

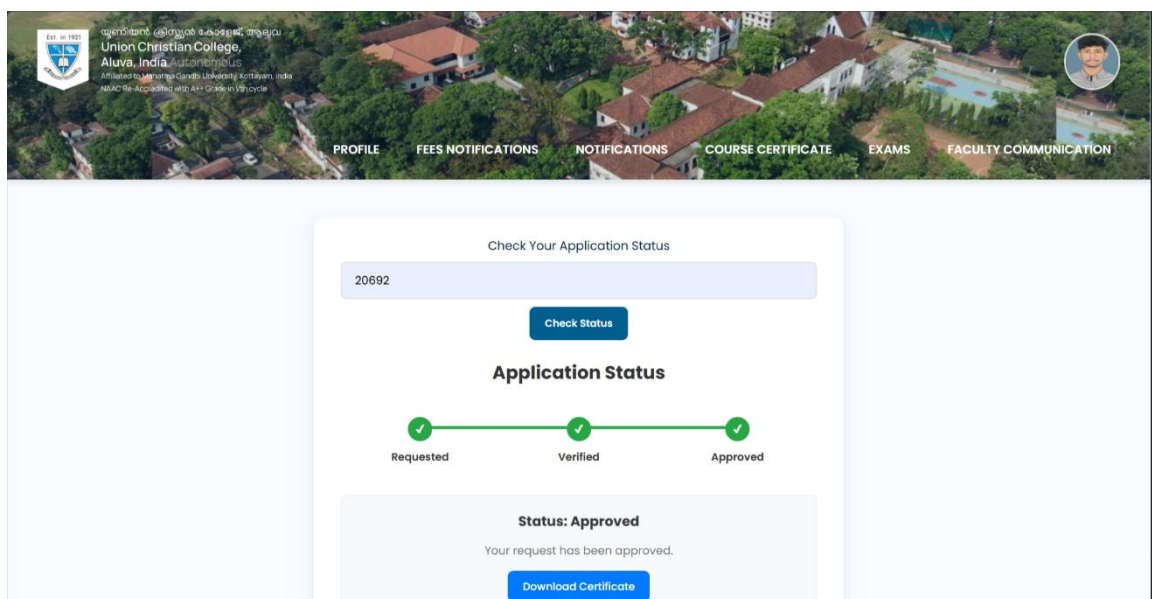
Application Status

Requested Verified Approved

Status: Approved
Your request has been approved.

Fig.4.3

11.Approved



Union Christian College, Aluva, India. Autonomous. Affiliated to Mar Thoma Syrian University, Kottayam, India. NAAC Re-Accredited with A++ Grade in Vth Cycle.

PROFILE FEES NOTIFICATIONS NOTIFICATIONS COURSE CERTIFICATE EXAMS FACULTY COMMUNICATION

Check Your Application Status

20692

Check Status

Application Status

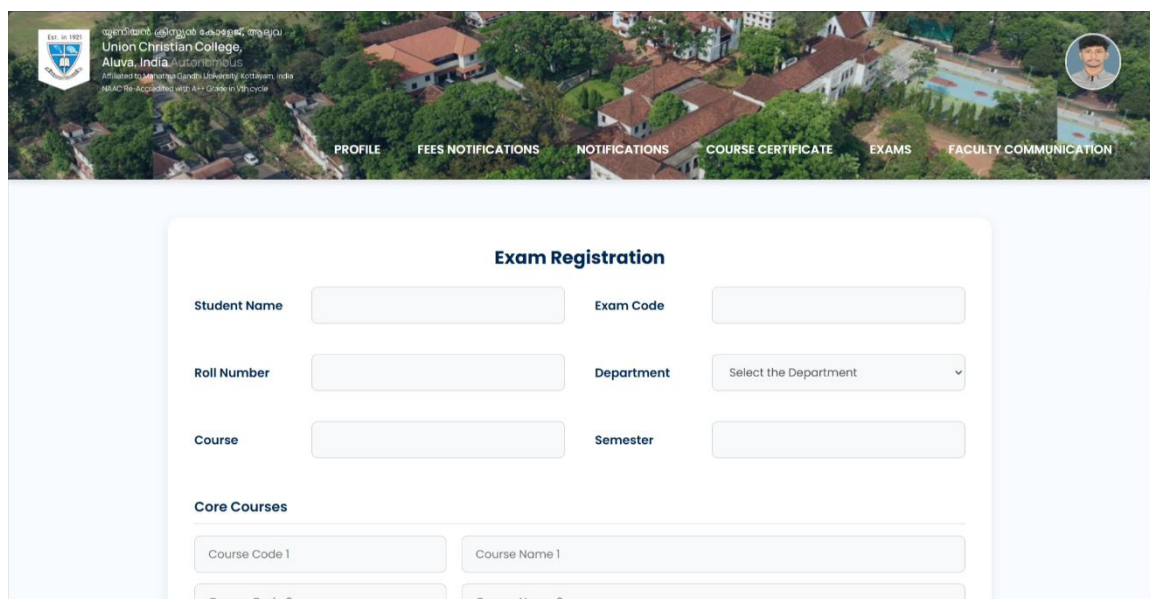
Requested Verified Approved

Status: Approved
Your request has been approved.

Download Certificate

Fig.4.4

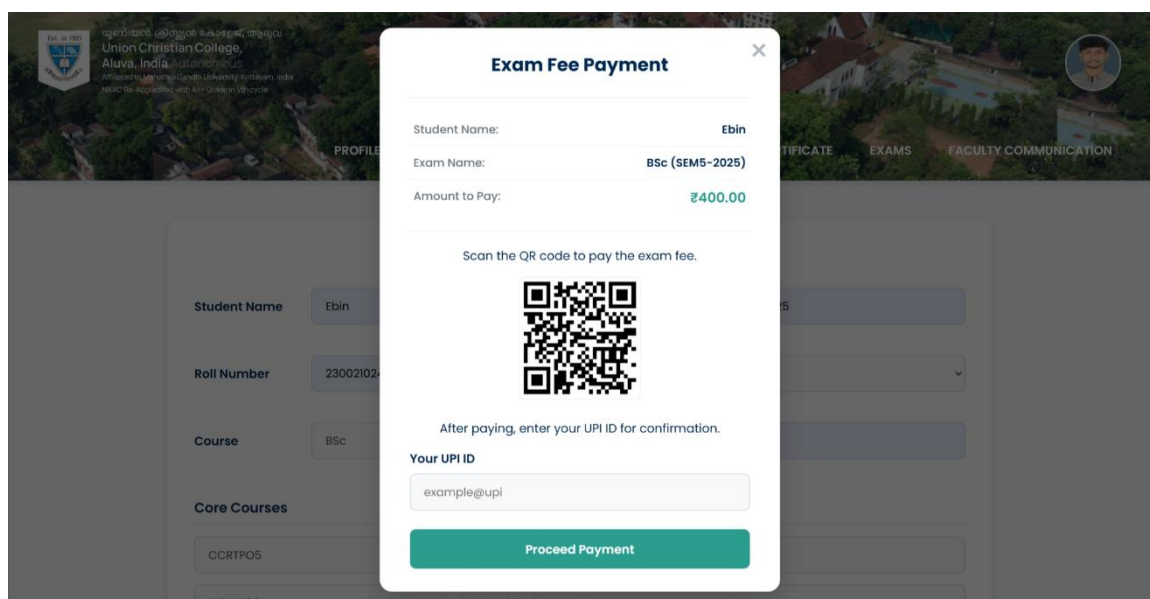
12.Exam Registration



The screenshot shows the 'Exam Registration' form within a web portal for Union Christian College, Aluva, India. The header includes the college logo, name, and affiliations, along with a navigation menu: PROFILE, FEES NOTIFICATIONS, NOTIFICATIONS, COURSE CERTIFICATE, EXAMS, and FACULTY COMMUNICATION. The form itself has a title 'Exam Registration' and contains several input fields: 'Student Name', 'Exam Code', 'Roll Number', 'Department' (a dropdown menu with 'Select the Department' as the placeholder), 'Course', and 'Semester'. Below these is a section for 'Core Courses' with two rows, each containing 'Course Code' and 'Course Name' input fields.

Fig.5

13.Exam Fee Payment



The screenshot displays the 'Exam Fee Payment' modal. The modal shows the student's details: 'Student Name: Ebin', 'Exam Name: BSc (SEM5-2025)', and 'Amount to Pay: ₹400.00'. It instructs the user to 'Scan the QR code to pay the exam fee.' and provides a QR code. Below the QR code, it says 'After paying, enter your UPI ID for confirmation.' and has a field for 'Your UPI ID' with the placeholder 'example@upi'. A green 'Proceed Payment' button is at the bottom. The background shows the same portal interface as Fig.5, but with the modal overlaid.

Fig.5.1

14.Payment Confirmation

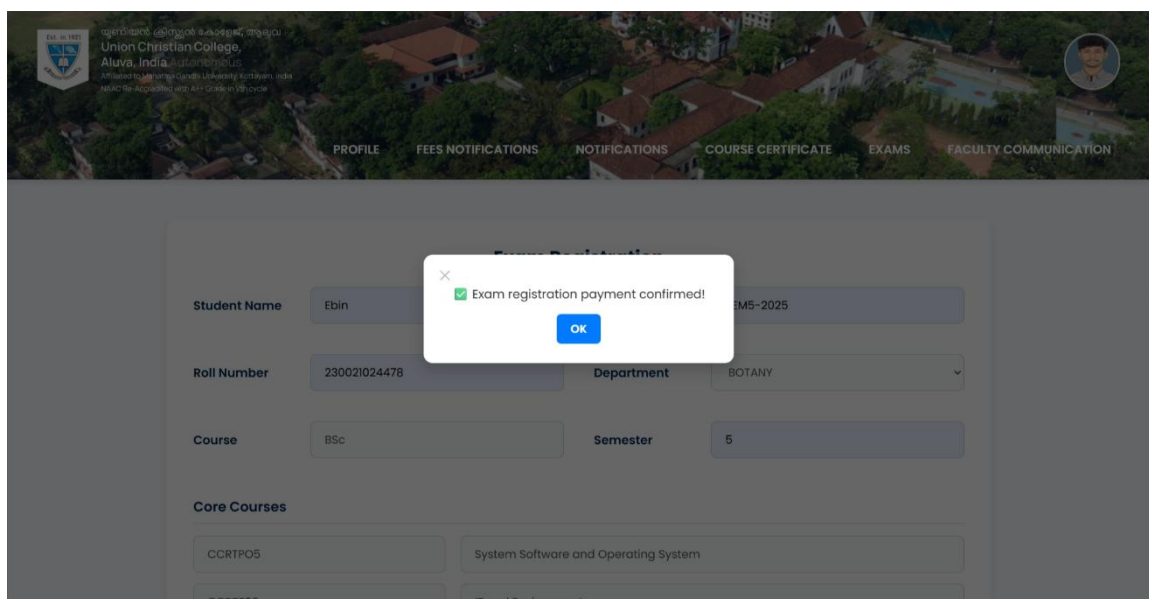


Fig.5.2

15.Faculty Communication

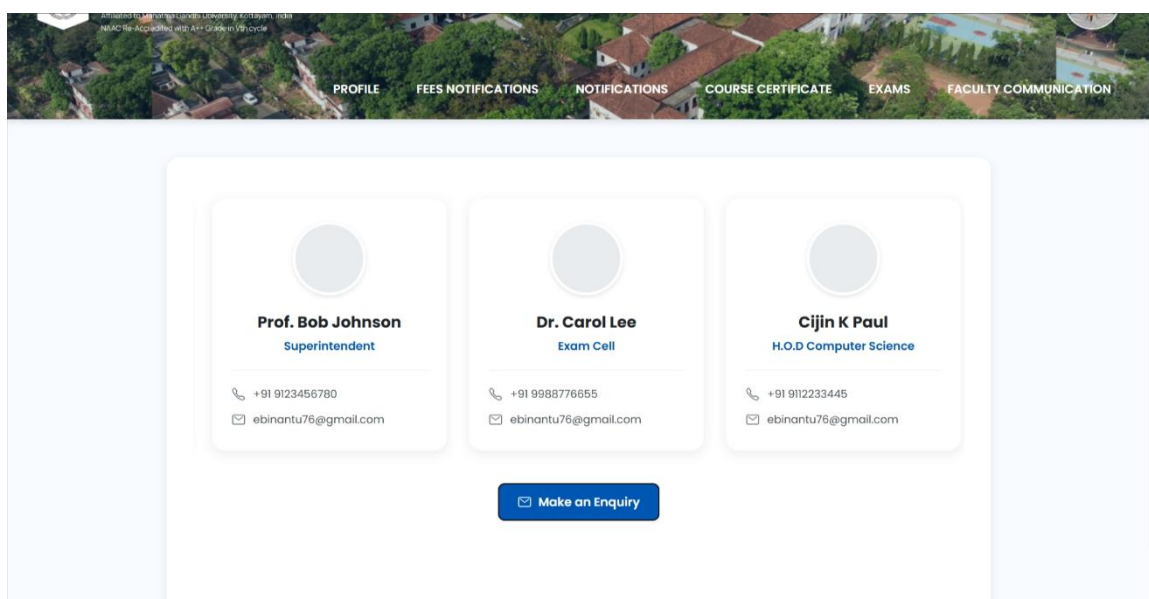


Fig.6

16. Contact Form

The image shows a contact form titled "Contact Us" centered on a light blue background. At the top of the form is a blue button with a close icon and the text "Close Form". Below the title, there are four input fields: "Department Email" (containing "name@example.com"), "Your Email" (empty), "Subject" (containing "How can we help?"), and "Message" (containing "Enter your message here..."). At the bottom of the form is a blue button with the text "Send Message" and a right-pointing arrow.

Fig.6.1

4. SYSTEM IMPLEMENTATION

Implementation is the stage in the project where the theoretical design is turned a working system. The most crucial stage is achieving a successful new system and giving the user confidence in that the new system will work efficiently and effectively in the implementation stage.

The stage consists of

- Testing the developed program with simple data.
- Detections and corrections.
- Creating whether the system is meets user requirements.
- Testing whether the system.
- Making necessary changes as desired by the user.
- Training user personnel

5.LIMITATIONS AND FUTURE ENHANCEMENTS

This system is very flexible so that the maintenance and further amendments based on the changing environment and requirements can be made easily. Any change that leads to the system failures is prevented with security measures.

Limitations:

- The portal depends entirely on internet connectivity. Students in areas with poor or no internet access may find it difficult to use the system effectively.
- Regular updates and maintenance are required to keep the portal functional and secure. Lack of technical support may lead to downtime or unresolved technical issues.
- Setting up the system, integrating databases, and training staff or students may require significant initial investment in terms of time and resources.
- The system may not fully meet the specific needs of every institution or department without further customization, which could require programming expertise.

- The effectiveness of the portal depends on the accuracy of the data entered. Any errors in input could lead to incorrect records or misinformation.
- As the number of users grows, the system may face performance or speed issues if not designed for scalability.

This project can be extended in future to provide additional facilities

- Introduce real-time analytics and reporting tools for better insights into student performance and payments.
- Expand notification services to include custom alerts and in-app push notifications.
- Implement multi-gateway payment integration for greater flexibility and transaction reliability.
- Add AI-based features such as performance tracking and fee reminder automation.
- Enable automated cloud backup to ensure continuous data protection.
- Provide multi-language support to cater to a wider range of users.
- Develop an admin dashboard for advanced monitoring and system control.
- This project done with detailed analysis of existing system and a careful design. So that future modification can be done in efficient manner with minimum disturbance to the system.

This project is mainly supported for multiuser environment. That is more than one user can use simultaneously. The system developed should be secured and protected against all possible hazards.

6.BIBLIOGRAPHY

Reference:

- 1.HTML - <https://developer.mozilla.org/en-US/docs/Web/HTML>
- 2.PHP - <https://developer.mozilla.org/en-US/docs/Glossary/PHP>
- 3.JAVASCRIPT - <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- 4.SQL, CSS - www.w3schools.com

Website reference:

1. www.w3schools.com
2. www.wikipedia.com
3. www.geminiai.com